

وَقُلْ رَبِّ ارْحَمْنِي عِلْمًا

سُبْحَانَكَ لَا عِلْمَ لَنَا إِلَّا مَا عَلَّمْتَنَا إِنَّكَ أَنْتَ الْعَلِيمُ الْحَكِيمُ

Dedications

In the name of Allah, the Most Gracious, the Most Merciful

To our beloved parents,

*whose unwavering support, endless sacrifices, and constant prayers
have been the foundation of our success.*

*To our teachers and mentors,
who guided us with knowledge and wisdom
throughout our academic journey.*

*To the team at Novation City,
who welcomed us with open arms and provided
invaluable learning opportunities.*

*To our families and friends,
whose encouragement and understanding
made this achievement possible.*

With deep gratitude and appreciation.

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Chapter 1

Conceptual Analysis

Introduction

This chapter presents the conceptual analysis of the current IoT REAP platform. Web-based learning and remote laboratory environment built around training paths, virtual machine sessions, hardware reservations, multimedia learning content, discussion features, and administrative infrastructure management. The chapter models the system through actor identification, requirements specification, use case analysis, sequence and activity modeling, class-level conceptualization, and the physical data model that supports platform execution.

1 Identification of the Actors

The platform is organized around role-based access control implemented in the application through the user roles `engineer`, `teacher`, and `admin`, in addition to the public unauthenticated state. Each actor interacts with the same platform from a different operational perspective, and each role is associated with a distinct set of services, permissions, and workflows.

Unauthenticated User (Guest) represents any visitor who has not yet authenticated in the system. This actor can browse public training paths, access legal and informational pages, consult public discussion threads in read-only mode, view training path reviews, verify public certificates, initiate authentication and registration workflows, and access public checkout result pages after external payment redirection. The guest actor therefore serves as the entry point into the platform ecosystem.

Engineer is the primary learner and remote laboratory user. This actor enrolls in training paths, consumes educational resources such as articles and videos, takes quizzes, tracks progress, writes notes, participates in discussions, posts reviews, claims certificates, and initiates payment and refund operations when a training path is monetized. Beyond the learning dimension, the engineer also provisions virtual machine sessions, opens remote desktop access through Guacamole, configures connection preferences, reserves USB devices and cameras, attaches hardware to active sessions, enters waiting queues when devices are busy, and interacts with session-bound lab resources.

Teacher is the content producer and pedagogical supervisor. This actor creates and manages training paths, modules, and training units; authors articles; uploads and manages videos and captions; creates and publishes quizzes; reviews learner interactions through analytics dashboards; moderates instructional discussions; submits training paths for administrative approval; and requests payout for generated revenue. The teacher role links educational design with operational control over learning content.

Administrator is the platform governance and infrastructure actor. This role supervises users, teacher approval, training path approval, featured training path ordering, payment-related review processes, payout validation, infrastructure supervision, Proxmox server and node management, VM assignment approval, hardware gateway verification, camera and reservation management, moderation of flagged forum content, maintenance operations, alerts, and activity logging. Administrators also have access to impersonation and compliance-sensitive actions such as suspension and GDPR-oriented user deletion.

Stripe (External Payment Processor) is an external system actor that supports the platform commerce workflows. IoT-REAP integrates with Stripe through hosted checkout redirections and signed webhook callbacks to confirm payments, create internal payment records, and react to asynchronous state changes. Stripe also participates in finance workflows such as refunds and instructor payouts, where administrators trigger payout transfers and refund executions through Stripe APIs while preserving auditability and transaction integrity.

The interaction among these actors reveals that the platform combines four complementary dimensions: e-learning delivery, instructor authoring, virtual laboratory provisioning, and infrastructure governance. This combination distinguishes the project from a conventional learning management system because educational workflows are directly connected to remote computing and physical IoT resource access.

2 Requirements Specifications

This section formalizes the system requirements derived from the implemented routes, services, and domain models. The requirements are divided into functional and non-functional categories in order to describe both what the platform must do and how well it must perform those responsibilities.

2.1 Functional Requirements

The platform must support the following functional requirements according to actor responsibilities and shared system behavior.

Unauthenticated User (Guest):

- Access account registration and login workflows
- Consult the landing page and public legal pages
- Browse public training paths and training path details
- Read public reviews and forum threads in read-only mode

- Verify and download publicly accessible certificates through verification links
- View public search results, trending queries, and category-based discovery
- Access checkout success and cancellation pages after external payment flows

Engineer:

- Enroll in and unenroll from training paths
- Access owned or enrolled training units and learning materials
- Read articles and track article completion
- Stream instructional videos and update playback progress
- Attempt quizzes, submit answers, and review attempt history
- Mark training units complete or incomplete based on learning progress
- Create, update, and delete personal notes for training units
- Create, edit, and delete reviews for completed or relevant training paths
- Participate in forum discussions by creating threads, replying, voting, and flagging content
- View notifications and manage read status
- Claim, verify, and download certificates
- Initiate checkout for paid training paths
- Review payment history and submit refund requests
- Provision, extend, view, and terminate virtual machine sessions
- Retrieve Guacamole access tokens for remote desktop connectivity
- Manage connection preference profiles for available protocols
- Browse available Proxmox virtual machines and snapshots
- Reserve USB devices and cameras
- Attach or detach session hardware and join hardware waiting queues
- Access session camera streams, request control, move PTZ-enabled cameras, and change stream resolution

Teacher:

- Create, edit, archive, restore, and delete training paths
- Organize training paths into modules and training units
- Reorder modules and training units
- Create and maintain quiz structures, questions, and publication status
- Create, update, and delete article content
- Upload, retry, and delete training videos
- Manage video captions
- Submit training paths for administrative review
- View teacher analytics for enrollment, earnings, and learning funnel data
- Access a teacher discussion inbox and moderate course-related threads
- Request revenue payout
- Submit training unit virtual machine assignment requests

Administrator:

- Access the administrative dashboard and KPI views
- Approve, reject, feature, unfeature, and order training paths
- Review pending VM assignment requests
- Manage reservations for devices and cameras
- Block or lock a USB device or camera to prevent new reservations (maintenance or administrative hold), or toggle an "open-for-all" mode to allow immediate non-reserved access
- Manage Proxmox servers and nodes
- Start, stop, reboot, and shut down virtual machines through node operations
- Register, verify, update, and remove gateway nodes
- Discover hardware devices and refresh infrastructure status
- Bind, unbind, attach, detach, dedicate, and configure devices and cameras

- Assign and unassign cameras to virtual machines
- Approve teacher accounts and manage user roles
- Suspend, unsuspend, impersonate, and delete user accounts where authorized
- Review and process refund requests
- Review, approve, reject, and process payout requests
- Monitor video processing status
- Place devices and cameras in maintenance mode
- Moderate flagged threads and replies
- Manage system alerts and consult activity logs

3 Use Case Diagram

After identifying actors and requirements, the conceptual analysis of the platform can be refined through use case modeling. The implemented application suggests one general use case view and several actor-specific refinements. The general model places the platform at the center and connects the four user actors (Guest, Engineer, Teacher, Administrator) as well as the external Stripe actor to the services they invoke: public discovery, learning progression, content authoring, infrastructure access, commerce, and administrative governance.

3.1 General Use Case Diagram

The general use case diagram groups system behavior into four principal subsystems:

- **Public Access and Discovery:** registration, authentication, public browsing, search, certificate verification
- **Learning and Participation:** enrollment, content consumption, quizzes, notes, reviews, forum participation, notifications, certificates
- **Teaching and Publishing:** training path authoring, multimedia management, quiz management, analytics, payout requests, submission for review
- **Remote Laboratory and Administration:** VM provisioning, remote access, reservation management, device control, infrastructure supervision, moderation, refunds, payouts, alerts, and logs

In this model, the Engineer extends the Guest by inheriting public consultation capabilities and adding authenticated learning and virtual laboratory operations. The Teacher extends the Engineer's authenticated access with content authoring and instructional management. The Administrator occupies the supervisory position by controlling both business governance and infrastructure operations. Stripe is modeled as a standalone external actor that interacts with the platform through checkout, webhook callbacks, refunds, and payout-related flows. Figure 1.1 presents the general use case diagram of the IoT-REAP platform, showing the four user actors plus Stripe and their principal interactions with the system subsystems.

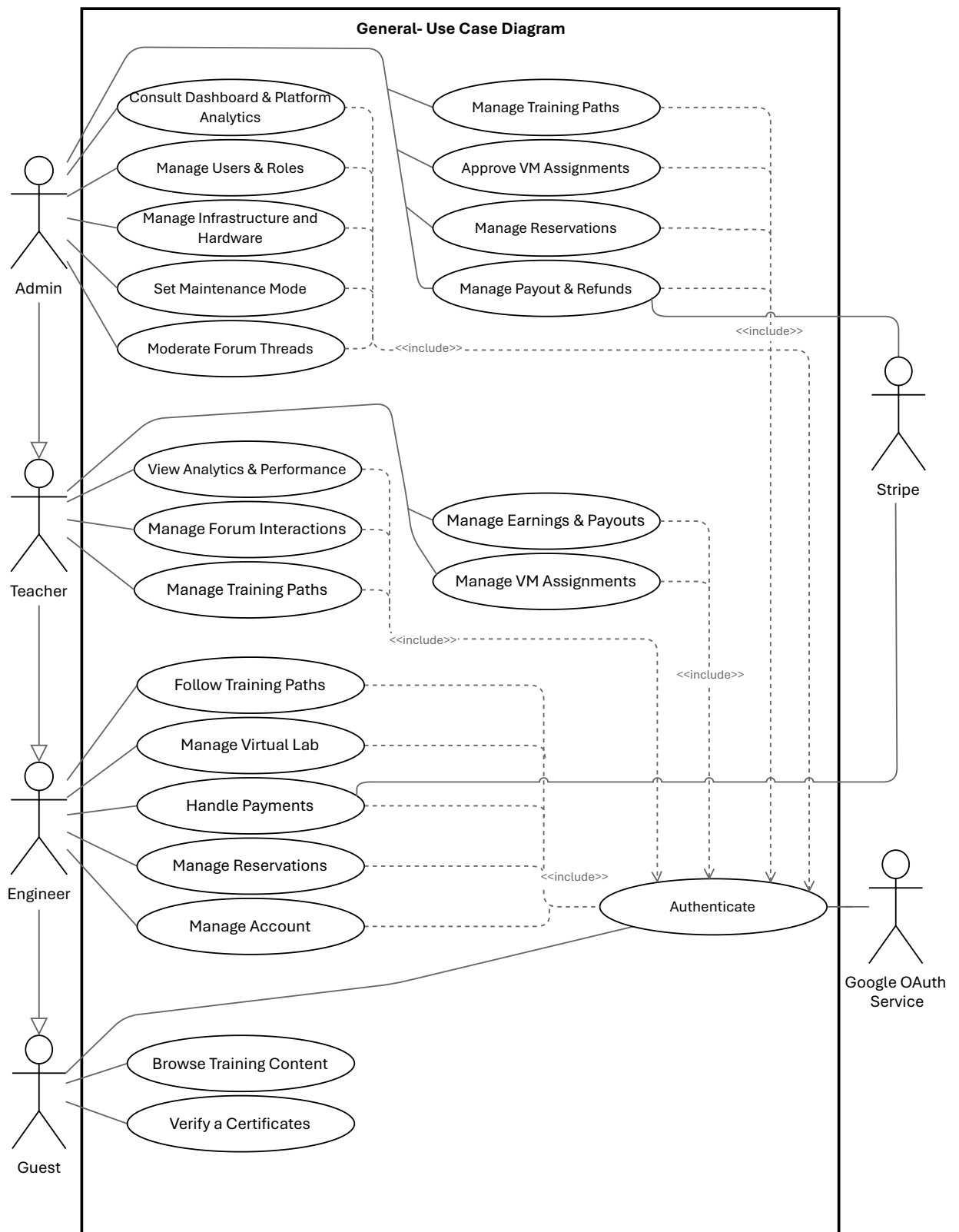


Figure 1.1: General Use Case Diagram

3.2 Refinement and Specification of Use Cases

The major use cases can be refined through representative specifications aligned with the actual implemented routes and domain behaviors.

3.2.1 Guest Use Cases

The Guest actor is restricted to public discovery and account access. The most important public use cases are account registration, training path consultation, certificate verification, and search.

3.2.1.1 Refinement of the Use Case Browse Training Content

Figure 1.2 presents the refinement of the Use Case Browse Training Content for the Guest actor. This use case includes the following sub-use cases:

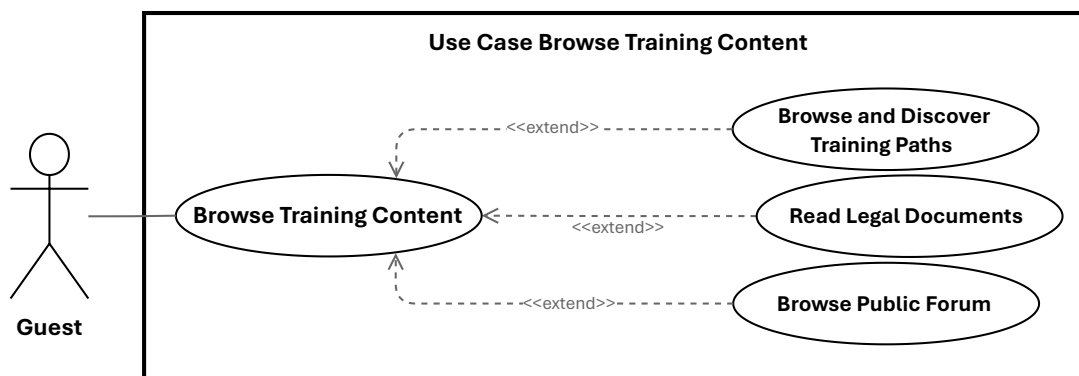


Figure 1.2: Use Case Browse Training Content

- **Specification of the Use Case Browse and Discover Training Paths**

Table 1.1: Use Case Specification: Browse and Discover Training Paths

Title	Browse and Discover Training Paths
Summary	Guest discovers available training paths by viewing the landing page, searching and filtering the catalog, and reading detailed path information including syllabus, reviews, pricing, and instructor details
Actor	Unauthenticated User (Guest)
Precondition	The platform is publicly accessible and at least one training path is published
Post-condition	Guest has viewed platform offerings and has enough information to decide whether to register
Basic Scenario	1. Guest arrives at the platform homepage; system displays featured training paths with titles, descriptions, and ratings 2. Guest navigates to the catalog; system lists all published training paths with categories, ratings, and pricing 3. Guest enters a keyword or selects category filters; system returns ranked matching results that may be sorted by price, rating, or enrollment count 4. Guest selects a training path; system displays the full detail page including description, learning objectives, module syllabus, pricing, duration, instructor profile, and enrolled-engineer statistics 5. Guest may read course reviews on the detail page to evaluate quality before deciding to register 6. If the guest attempts to enroll, the system redirects to the registration page
Error Scenario	1. If no featured paths are configured, the landing page displays a default welcome message 2. If no paths match the search criteria, the system displays a no-results message with suggestions 3. If the search service is unavailable, the system falls back to a static listing with a retry option 4. If a selected training path is not found or unpublished, the system redirects to the catalog

• **Specification of the Use Case Read Legal Documents**

Table 1.2: Use Case Specification: Read Legal Documents

Title	Read Legal Documents
Summary	Any user reads the platform's Terms of Service or Privacy Policy to understand usage rights and data handling practices
Actor	Unauthenticated User (Guest), Engineer, Teacher, Administrator
Precondition	The requested legal document is available and accessible
Post-condition	User views the complete and latest version of the selected legal document
Basic Scenario	1. User navigates to the legal section via footer links 2. User selects either Terms of Service or Privacy Policy 3. System retrieves the latest version of the selected document 4. System renders the document in a readable, properly formatted view 5. User scrolls through and reads the complete document
Error Scenario	1. If the document is unavailable or fails to load, the system displays a friendly error message with a retry option

- **Specification of the Use Case Browse Public Forum**

Table 1.3: Use Case Specification: Browse Public Forum

Title	Browse Public Forum
Summary	Guest browses and reads public forum threads and their replies to follow community discussions without authentication
Actor	Unauthenticated User (Guest)
Precondition	The platform is publicly accessible and the forum contains at least one published thread
Post-condition	Guest has read one or more forum threads and their approved replies
Basic Scenario	1. Guest navigates to the public forum page; system displays a paginated list of threads with titles, authors, and reply counts 2. Guest may sort threads by date, popularity, or category, or search by keyword 3. Guest selects a thread; system displays the full detail page showing the original post with author and timestamp, followed by all approved replies in chronological order 4. Guest reads the discussion and may view engagement metrics 5. If the guest attempts to post or reply, the system redirects to the registration page
Error Scenario	1. If no threads match a search query, the system displays a no-results message 2. If a selected thread is not found or has been removed, the system displays a not-found notice and returns to the listing

3.2.1.2 Refinement of the Use Case Authenticate

Figure 1.3 presents the refinement of the Use Case Authenticate for the Guest actor. This use case includes the following sub-use cases:

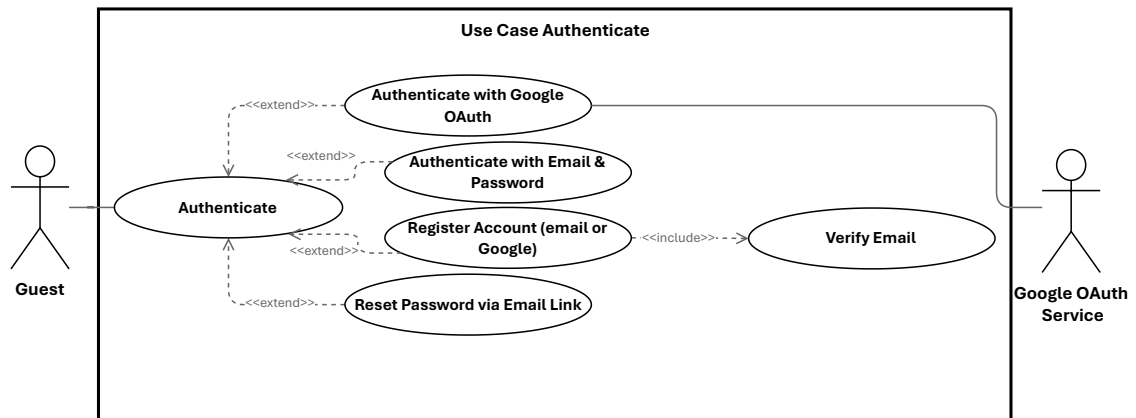


Figure 1.3: Use Case Authenticate

- **Specification of the Use Case Authenticate with Google OAuth**

Table 1.4: Use Case Specification: Authenticate with Google OAuth

Title	Authenticate with Google OAuth
Summary	Guest authenticates to the platform using a Google account through the OAuth 2.0 protocol, enabling single sign-on without manual credential entry
Primary Actor	Unauthenticated User (Guest)
Secondary Actor	Google OAuth Service
Precondition	The guest has a valid Google account; the Google OAuth integration is configured and active
Post-condition	The guest is authenticated and redirected to the role-appropriate dashboard, or prompted to select a role if a new account
Basic Scenario	1. Guest clicks the Google Sign-In button on the login or registration page 2. System redirects the guest to the Google OAuth consent screen 3. Guest authorizes the application and Google returns an authorization code 4. System exchanges the authorization code for user profile information 5. If the Google account matches an existing user, the system logs the user in 6. If the Google account is new, the system creates a pending account and presents the role selection page 7. Guest selects a role (Engineer or Teacher) and completes the signup 8. System authenticates the user and redirects to the appropriate dashboard
Error Scenario	1. If the Google OAuth callback fails, the system redirects to the login page with an error message 2. If the guest denies consent on the Google screen, the system returns to the login page 3. If the Google account email is already linked to a password-based account, the system prompts the user to log in with email and password

• Specification of the Use Case Authenticate with Email & Password

Table 1.5: Use Case Specification: Authenticate with Email & Password

Title	Authenticate with Email & Password
Summary	Guest authenticates to the platform using email and password to access protected features
Actor	Guest
Precondition	The guest has a registered and verified account
Post-condition	The guest is authenticated and receives a session with their assigned role
Basic Scenario	1. Guest navigates to the login page 2. System displays the login form with email and password fields and a Google OAuth button 3. Guest enters their credentials and submits the form 4. System validates the credentials against the stored hash 5. System creates an authenticated session with the user's role 6. System redirects the guest to their role-specific dashboard
Error Scenario	1. If the credentials are invalid, the system displays an error and increments the failed attempt counter 2. If the account is locked due to excessive failed attempts, the system prompts the guest to reset their password 3. If Google OAuth fails, the system falls back to the email/password login form

- **Specification of the Use Case Register Account (email or Google)**

Table 1.6: Use Case Specification: Register Account (email or Google)

Title	Register Account (email or Google)
Actor	Guest
Precondition	The guest is not authenticated and the platform allows new registrations
Post-condition	A new user account is created with pending email verification status
Basic Scenario	1. Guest navigates to the registration page 2. System displays the registration form with required fields 3. Guest enters their name, email address, and password 4. System validates the input fields and checks for duplicate email 5. System creates the account with pending verification status 6. System sends a verification email to the provided address 7. Guest clicks the verification link to activate the account
Error Scenario	1. If the email is already registered, the system prompts the guest to log in instead 2. If the password does not meet security requirements, the system rejects the submission 3. If the verification email expires, the system allows resending a new link

- **Specification of the Use Case Reset Password via Email Link**

Table 1.7: Use Case Specification: Reset Password via Email Link

Title	Reset Password via Email Link
Summary	Guest, engineer, or teacher resets their password via email link
Actor	Unauthenticated User (Guest), Engineer, Teacher
Precondition	The user has an active account with a registered email
Post-condition	The user's password is updated and they can log in with the new password
Basic Scenario	1. User clicks "Forgot Password" on the login page 2. User enters their email address 3. System validates the email and creates a password reset token 4. System sends a password reset email with a secure link 5. User clicks the link and is redirected to the reset form 6. User enters and confirms a new password 7. System validates the password meets security requirements 8. System updates the password and invalidates the reset token 9. System notifies the user via email that the password was changed
Error Scenario	1. If the email is not registered, the system displays a generic success message (security) 2. If the reset token expires, the system prompts the user to request a new one 3. If the new password does not meet requirements, validation fails

- **Specification of the Use Case Verify Email**

Table 1.8: Use Case Specification: Verify Email

Title	Verify Email
Summary	Guest, engineer, or teacher verifies their email address to activate their account
Actor	Unauthenticated User (Guest), Engineer, Teacher
Precondition	The user has registered an account but not verified their email
Post-condition	The user's email is marked as verified and full account access is granted
Basic Scenario	1. User registers a new account or logs in with an unverified email 2. System creates an email verification token 3. System sends a verification email with a secure link 4. User clicks the link and is redirected to the verification endpoint 5. System validates the token and marks the email as verified 6. System notifies the user that verification was successful 7. User is redirected to their dashboard with full access
Error Scenario	1. If the token expires, the system prompts the user to request a new one 2. If the token is invalid, the system displays an error message 3. If the email is already verified, the system notifies the user

3.2.1.3 Refinement of the Use Case Verify a certificate

Figure ?? presents the refinement of the Use Case Verify a certificate for the Guest actor. This use case includes the following sub-use cases:

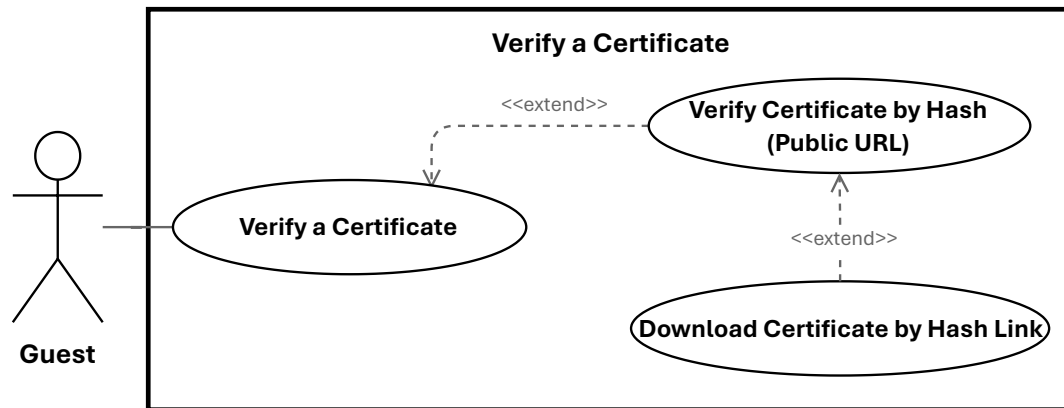


Figure 1.4: Use Case Verify a certificate

• Specification of the Use Case Verify a Certificate

Table 1.9: Use Case Specification: Verify Certificate

Title	Verify Certificate
Summary	Guest verifies the authenticity of a digital certificate using its hash
Actor	Unauthenticated User (Guest)
Precondition	The certificate exists and has been issued by the platform
Post-condition	The guest receives confirmation of the certificate's validity and details
Basic Scenario	1. Guest receives a certificate (e.g., via email or LinkedIn) 2. Guest navigates to the certificate verification page 3. Guest enters the certificate hash (or clicks a verification link) 4. System validates the hash and retrieves the certificate record 5. System displays the certificate details: recipient name, issue date, and credential title 6. System confirms the certificate is authentic and not revoked
Error Scenario	1. If the hash is invalid, the system displays an error message 2. If the certificate is revoked, the system notifies the guest 3. If the certificate has expired, the system displays an expiration warning

• Download Certificate by Hash Link**Table 1.10:** Use Case Specification: Download Certificate by Hash Link

Title	Download Certificate by Hash Link
Summary	Guest downloads a digital certificate using a verification hash link without authentication
Actor	Unauthenticated User (Guest)
Precondition	The certificate exists, is valid, and has not been revoked
Post-condition	The guest downloads the certificate file in PDF or image format
Basic Scenario	1. Guest receives a certificate download link via email or external platform 2. Guest clicks the download link containing the certificate hash 3. System validates the hash and retrieves the certificate record 4. System confirms the certificate is valid and not revoked 5. System generates and serves the certificate file for download 6. Guest saves the certificate file to their device
Error Scenario	1. If the hash is invalid, the system displays an error message 2. If the certificate is revoked, the system notifies the guest 3. If the certificate has expired, the system displays an expiration warning 4. If the download fails, the system offers a retry option

3.2.2 Engineer Use Cases

The Engineer actor focuses on training path enrollment, virtual lab access, and hands-on learning. The key use cases include training path enrollment and management, content consumption (videos, articles, quizzes), virtual machine session management, hardware device and camera control, reservation management, forum participation, certificate claims, payment operations, and account management.

3.2.2.1 Refinement of the Use Case Follow Training Paths

Figure 1.5 presents the refinement of the Use Case Follow Training Paths for the Engineer actor. This use case includes the following sub-use cases:

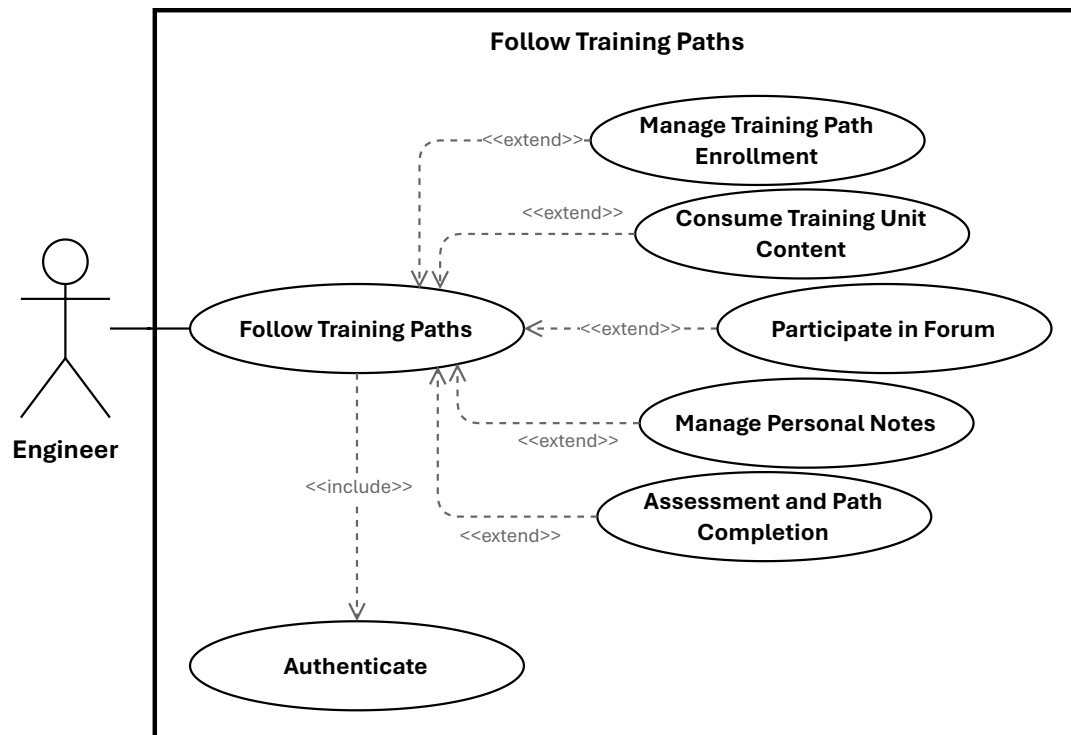


Figure 1.5: Use Case Follow Training Paths

- **Specification of the Use Case Manage Training Path Enrollment**

Table 1.11: Use Case Specification: Manage Training Path Enrollment

Title	Manage Training Path Enrollment
Summary	Engineer browses their enrolled training paths, enrolls in new ones, and unenrolls from paths they no longer wish to pursue
Actor	Engineer
Precondition	The engineer is authenticated and the target training path is published and available where applicable
Post-condition	The engineer's enrollment state is created or removed and access to training path content is granted or revoked accordingly
Basic Scenario	1. Engineer opens the dashboard; system displays all enrolled training paths with title, thumbnail, progress percentage, and status; engineer may filter by status or category 2. For Enroll: engineer selects a training path from the catalog and clicks Enroll; if the path is free enrollment is immediate; if paid the engineer is redirected to checkout; upon success the system creates the enrollment record and grants access to all modules and units 3. For Unenroll: engineer selects an enrolled training path and confirms the unenrollment action; system removes the enrollment record, revokes content access, and handles any applicable refund according to platform policy
Error Scenario	1. If the engineer has no enrollments, system displays an empty state with discovery suggestions 2. If the engineer is already enrolled, system displays a message and prevents duplicate enrollment 3. If the training path is unpublished or archived, enrollment is blocked and the path is marked accordingly 4. If payment fails, enrollment is not completed 5. If unenrollment is restricted due to completion status, system displays an explanatory message

• **Specification of the Use Case Consume Training Unit Content**

Table 1.12: Use Case Specification: Consume Training Unit Content

Title	Consume Training Unit Content
Summary	Engineer watches video lessons, reads article lessons, and marks training units complete or incomplete, with progress tracked and persisted throughout
Actor	Engineer
Precondition	The engineer is enrolled in the training path and the target training unit is accessible and published
Post-condition	The engineer's content progress is recorded and the overall training path progress is recalculated accordingly
Basic Scenario	1. Engineer opens a training unit from their enrolled training path 2. For Video: system loads the video player with the processed stream URL; engineer plays, pauses, seeks, or adjusts playback speed; system periodically records playback position; upon reaching the completion threshold the video is marked as watched 3. For Article: system renders the article with estimated reading time; engineer scrolls through the content; system tracks progress via scroll position or explicit confirmation; upon reaching the end the article is marked as read 4. Engineer may manually mark the unit complete or incomplete at any time; system persists the status and recalculates module and path-level progress percentages 5. Updated progress is reflected on the training path overview
Error Scenario	1. If the video has not finished processing, system displays a pending state with a retry option 2. If the video stream URL is unavailable, system shows an error with a fallback message 3. If the article is unpublished or fails to load, system shows a notice and prevents access 4. If progress saving fails, system retries silently and preserves the last known state 5. If the engineer is not enrolled, system denies access and progress updates

• Specification of the Use Case Participate in Forum**Table 1.13:** Use Case Specification: Participate in Forum

Title	Participate in Forum
Summary	Engineer engages with training path discussion forums by posting threads or replies, upvoting helpful content, and flagging inappropriate content for moderation
Actor	Engineer
Precondition	The engineer is enrolled in at least one training path with an active forum
Post-condition	The engineer's post, upvote, or flag is persisted and reflected in the forum accordingly
Basic Scenario	1. Engineer navigates to the forum section of an enrolled training path 2. For Post: engineer creates a new discussion thread or replies to an existing post; system validates content, persists the post, and notifies subscribed participants 3. For Upvote: engineer clicks the upvote button on a thread or reply; system increments the upvote count and records the vote; engineer may remove their upvote at any time 4. For Flag: engineer clicks the flag button to report inappropriate content; system records the flag and may hide the content pending moderator review; engineer may remove their flag if desired
Error Scenario	1. If the forum is closed or locked, system prevents new posts and replies 2. If content validation fails, system displays specific error messages and preserves input 3. If the engineer has already upvoted, system prevents duplicate votes 4. If the content has already been flagged, system acknowledges the existing report

• Specification of the Use Case Manage Personal Notes

Table 1.14: Use Case Specification: Manage Personal Notes

Title	Manage Personal Notes
Summary	Engineer creates, edits, and deletes private notes attached to training units for personal reference
Actor	Engineer
Precondition	The engineer is enrolled in the training path and the training unit is accessible
Post-condition	The note is persisted, updated, or removed; notes remain private and are never visible to other users
Basic Scenario	1. Engineer opens a training unit; system displays any existing notes or an empty editor 2. Engineer writes or edits note content; system validates and persists the note on save 3. Engineer may delete a note at any time; system requests confirmation then removes it
Error Scenario	1. If note content exceeds the maximum length, validation fails with a specific message 2. If the engineer is not enrolled, system prevents note creation 3. If save or deletion fails, system preserves the current state and logs the error

- **Specification of the Use Case Assessment and Path Completion**

Table 1.15: Use Case Specification: Assessment and Path Completion

Title	Assessment and Path Completion
Summary	Engineer takes quiz assessments to test knowledge, writes a course review to share feedback, and claims a completion certificate upon finishing all training path requirements
Actor	Engineer
Precondition	The engineer is enrolled in the training path; the quiz is published for assessment; all modules and quizzes are completed before certificate claim
Post-condition	Quiz attempt is recorded and scored; review is persisted and visible after approval; certificate is generated and available on the engineer's profile
Basic Scenario	1. For Quiz: engineer opens a training unit containing a quiz; system displays instructions, question count, and time limit; engineer answers all questions and submits or time expires; system grades the attempt, displays results with correct answers and explanations, and updates training unit progress based on passing criteria 2. For Review: engineer navigates to the training path review section; selects a star rating and optionally writes a text review; system validates content, persists the review, recalculates the path average rating, and makes the review visible once approved; if a review already exists the system offers an edit option instead 3. For Certificate: engineer opens the completed training path overview; system verifies all completion requirements are met, generates a unique certificate with a verification hash, and makes it available for download, sharing, and display on the engineer's public profile
Error Scenario	1. If the quiz has already been attempted and retakes are not allowed, access is blocked 2. If quiz submission fails, system preserves answers and allows retry 3. If the rating is missing, system rejects the review submission 4. If review text violates content policies, system flags it for moderation 5. If completion requirements are not fully met, system displays the remaining items and blocks certificate generation 6. If certificate generation fails, system logs the error and allows retry

3.2.2.2 Refinement of the Use Case Manage Reservations

Figure 1.6 presents the refinement of the Use Case Manage Reservations for the Engineer actor. This use case includes the following sub-use cases:

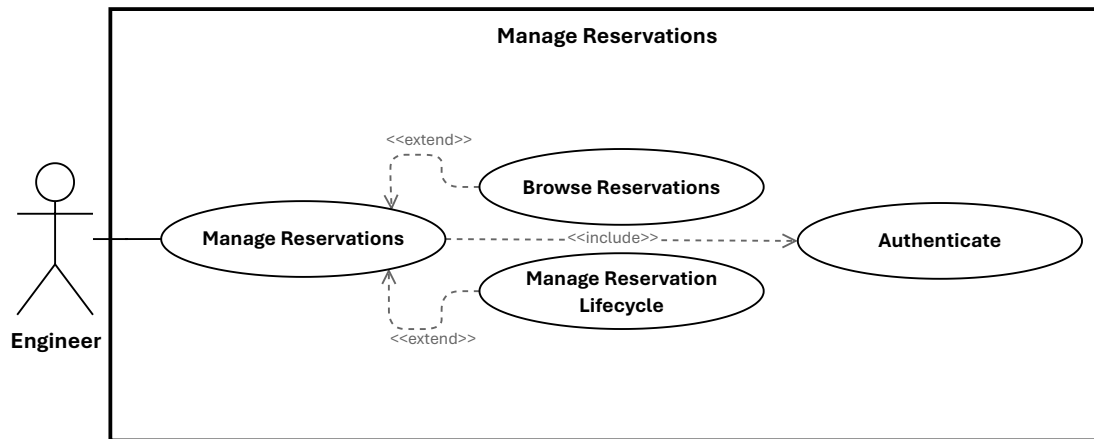


Figure 1.6: Use Case Manage Reservations

- Specification of the Use Case Browse Reservations

Table 1.16: Use Case Specification: Browse Reservations

Title	Browse Reservations
Summary	Engineer views all their reservations for USB devices, cameras, and VM resources, and inspects the full details of any individual reservation
Actor	Engineer
Precondition	The engineer is authenticated and has created at least one reservation
Post-condition	Engineer has viewed their reservation list and the complete details of a selected reservation including dates, status, resource type, and any pending actions
Basic Scenario	1. Engineer navigates to the reservations section; system retrieves and displays all reservations with resource type, date range, and status; engineer may filter by resource type or status 2. Engineer selects a reservation; system displays the full detail view showing resource type, date range, status, special requirements, approval status, and any available actions
Error Scenario	1. If no reservations exist, system displays an empty state with an option to create one 2. If a selected reservation has been deleted, system shows a not-found notice and returns to the list

- **Specification of the Use Case Manage Reservation Lifecycle**

Table 1.17: Use Case Specification: Manage Reservation Lifecycle

Title	Manage Reservation Lifecycle
Summary	Engineer consults the availability calendar, creates new reservations for USB devices, cameras, or VM resources, and cancels existing ones when no longer needed
Actor	Engineer
Precondition	The engineer is authenticated and has access to the reservation system; the desired resource must be available for new reservations
Post-condition	A reservation is created and pending approval, or an existing reservation is cancelled and the resource is released for other users
Basic Scenario	1. Engineer opens the reservation calendar; system displays a calendar view with resource availability and existing reservations shown as blocked time slots; engineer may navigate between dates to find a suitable slot 2. For Create: engineer selects an available time slot and resource type, system checks for scheduling conflicts, engineer submits the request, system creates the reservation with pending status and confirms the submission with approval status 3. For Cancel: engineer selects an existing reservation and confirms cancellation; system cancels the record, releases the resource, and makes it available for other reservations
Error Scenario	1. If calendar data fails to load, system shows an error with a retry option 2. If the selected time slot conflicts with an existing reservation, system shows alternative availability 3. If the resource is not available, system displays availability information 4. If the reservation has already started, cancellation may be restricted 5. If cancellation fails, system logs the error and preserves the reservation

3.2.2.3 Refinement of the Use Case Manage Virtual Lab

Figure 1.7 presents the refinement of the Use Case Manage Virtual Lab for the Engineer actor. This use case includes the following sub-use cases:

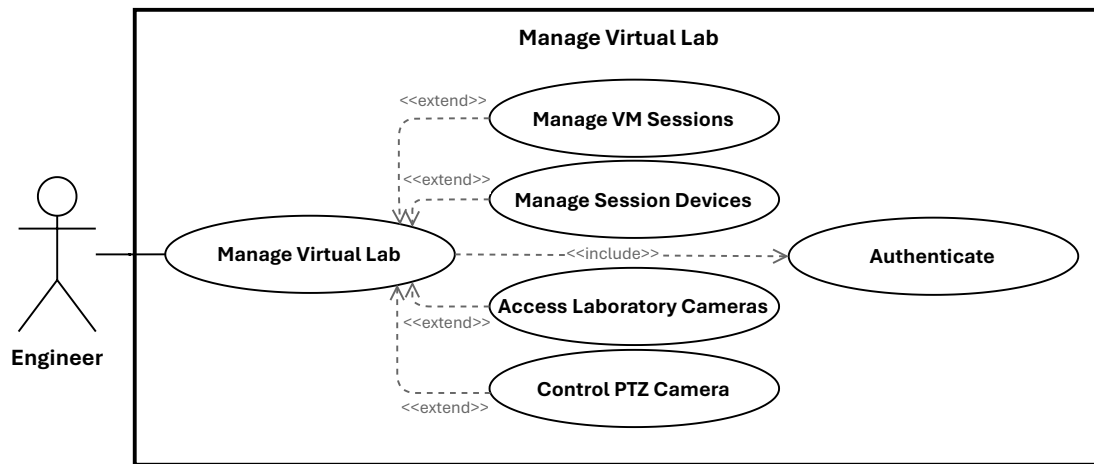


Figure 1.7: Use Case Manage Virtual Lab

- **Specification of the Use Case Manage VM Sessions**

Table 1.18: Use Case Specification: Manage VM Sessions

Title	Manage VM Sessions
Summary	Engineer views, provisions, extends, and terminates virtual machine sessions to access and manage their remote laboratory environment
Actor	Engineer (and Administrator when authorized)
Precondition	The engineer is authenticated, verified, and authorized through the <code>provision-vm</code> ability where applicable
Post-condition	The VM session is created, extended, or terminated and infrastructure resources are allocated or released accordingly
Basic Scenario	1. Engineer opens the VM sessions interface; system displays all sessions with status (active, expired, terminated), creation time, and remaining duration; engineer may filter by status or date range 2. For Provision: engineer submits a request to create a new session; system validates authorization and rate-limit constraints, selects suitable server and node resources, provisions the session record with associated infrastructure metadata, and returns the created session to the interface 3. For Extend: engineer selects an active session and chooses an extension duration; system validates eligibility and rate limits, updates the expiration time, and confirms the new deadline 4. For Terminate: engineer confirms the termination of an active session; system validates the request, releases associated infrastructure resources, and marks the session as terminated
Error Scenario	1. If no sessions exist, system displays an empty state with an option to provision one 2. Authorization failure or rate-limit excess prevents provisioning and surfaces an explanatory error 3. If no infrastructure capacity is available, system returns a controlled failure 4. Extension is blocked if the session has expired or the maximum extension limit is reached 5. If termination fails, system logs the error and preserves the session state 6. Provisioning or extension failures preserve traceability through logs or alerts

- **Specification of the Use Case Manage Session Devices**

Table 1.19: Use Case Specification: Manage Session Devices

Title	Manage Session Devices
Summary	Engineer attaches and detaches USB devices to and from an active VM session, and joins a waiting queue when a desired device is currently in use
Actor	Engineer
Precondition	The engineer has an active VM session
Post-condition	The device is attached to or detached from the session, or the engineer is queued and notified when the device becomes available
Basic Scenario	1. Engineer views their active session; system displays available devices with availability status 2. For Attach: engineer selects an available device; system validates compatibility and availability, establishes the connection between the device and the VM, and confirms attachment; the device becomes accessible from within the VM environment 3. For Detach: engineer selects an attached device and confirms detachment; system disconnects the device, releases it for use by other sessions, and confirms the action 4. For Queue: if the desired device is in use, engineer selects the option to join the waiting queue; system adds the engineer with their queue position and estimated wait time, and notifies them when the device becomes available
Error Scenario	1. If the device is already in use, system displays availability information and offers the queue option 2. If the device is incompatible with the VM, system prevents attachment 3. If attachment or detachment fails, system logs the error and allows retry 4. If detachment is requested while a process is actively using the device, system warns or delays the action 5. If the engineer is already in the queue, system displays their current position 6. If the queue is full, system prevents joining and displays capacity information

• Specification of the Use Case Access Laboratory Cameras

Table 1.20: Use Case Specification: Access Laboratory Cameras

Title	Access Laboratory Cameras
Summary	Engineer browses available laboratory cameras, views their live streams, and adjusts stream resolution to monitor laboratory equipment during an active VM session
Actor	Engineer
Precondition	The engineer has an active VM session and at least one camera is configured for the laboratory
Post-condition	Engineer is viewing a live camera stream at the selected resolution
Basic Scenario	1. Engineer opens the camera panel for their active session; system displays available cameras with status (available, in use, offline) and capabilities (PTZ support, resolution options) 2. Engineer selects a camera; system displays available resolution options with details (width, height, frame rate, bandwidth requirements) 3. Engineer selects a resolution; system initiates the stream connection and displays the live video feed in a player window 4. Engineer may switch resolution at any time while viewing; system reconfigures the stream and updates bandwidth requirements accordingly 5. Stream continues until the engineer stops viewing or the session ends
Error Scenario	1. If no cameras are available or configured, system displays an empty state 2. If camera status cannot be retrieved, system shows an error with a retry option 3. If a camera is offline, system displays an unavailable message 4. If the stream fails to load, system offers retry and may suggest a lower resolution 5. If resolution switching fails, system preserves the current resolution

• Specification of the Use Case Control PTZ Camera

Table 1.21: Use Case Specification: Control PTZ Camera

Title	Control PTZ Camera
Summary	Engineer acquires exclusive control over a PTZ camera, sends pan, tilt, and zoom commands to adjust its view, and releases control when finished
Actor	Engineer
Precondition	The engineer has an active VM session and is viewing a camera that supports PTZ control
Post-condition	Camera control is acquired, commands are applied, and control is released and made available to other users
Basic Scenario	1. Engineer views a PTZ-capable camera stream; system displays control availability status 2. Engineer requests camera control; system checks availability and if free grants exclusive control to the engineer, updates control status, and notifies other users 3. Engineer uses the PTZ control interface to send pan (horizontal), tilt (vertical), and zoom commands; system forwards each command to the camera, which moves to the new position; stream updates to reflect the new view 4. When finished, engineer releases control; system revokes exclusive access, marks the camera as available, and notifies other users
Error Scenario	1. If the camera does not support PTZ, system disables control options 2. If the camera is already controlled by another user, system displays a queue option 3. If a PTZ command is out of range or invalid, system ignores or limits it 4. If the camera does not respond to a command, system displays an error 5. If control release fails, system logs the error and preserves the current control state

3.2.2.4 Refinement of the Use Case Handle Payments

Figure 1.8 presents the refinement of the Use Case Handle Payments for the Engineer actor. This use case includes the following sub-use cases:

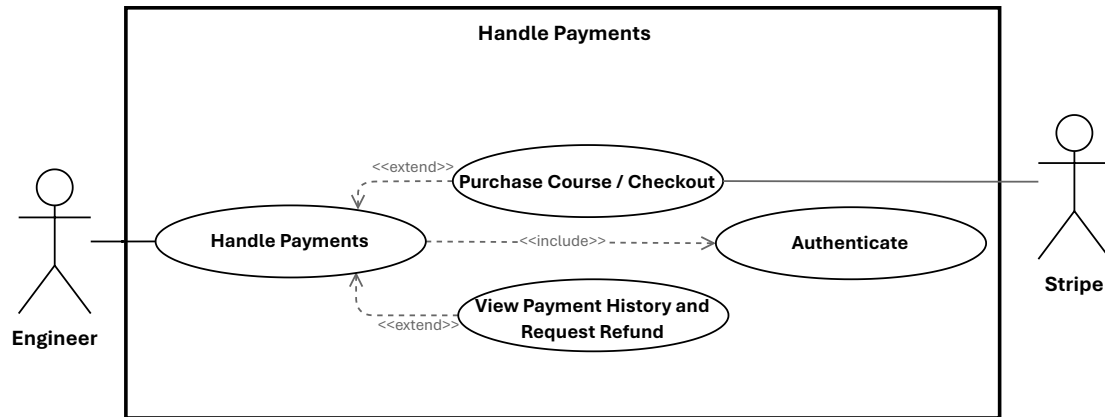


Figure 1.8: Use Case Handle Payments

- **Specification of the Use Case Purchase Course / Checkout**

Table 1.22: Use Case Specification: Purchase Course / Checkout

Title	Purchase Course / Checkout
Summary	Engineer purchases a paid training path through Stripe checkout and is directed to a confirmation or cancellation screen depending on the outcome
Primary Actor	Engineer
Secondary Actor	Stripe
Precondition	The engineer has selected a training path for enrollment and has a valid payment method
Post-condition	Payment is confirmed and enrollment is activated, or the checkout is cancelled and no enrollment or payment record is created
Basic Scenario	1. Engineer reviews the training path details and price, then initiates checkout 2. System redirects the engineer to the Stripe payment page 3. For Success: engineer enters payment details and confirms; Stripe processes the payment and sends a webhook confirmation; system activates the enrollment, generates a receipt, and displays a confirmation screen with transaction details and a link to the training path 4. For Cancellation: engineer aborts on the Stripe page; Stripe redirects to the platform cancellation screen; system displays a cancellation notice and offers options to retry checkout or return to the training path details; no payment or enrollment record is created
Error Scenario	1. If the payment is declined, system displays the rejection reason 2. If the webhook is not received, system polls Stripe for status 3. If the training path price changed during checkout, system requests re-confirmation 4. If enrollment activation is delayed after successful payment, system shows a processing message 5. If the Stripe redirect fails in either direction, engineer may need to manually navigate back to the platform

• **Specification of the Use Case View Payment History and Request Refund**

Table 1.23: Use Case Specification: View Payment History and Request Refund

Title	View Payment History and Request Refund
Summary	Engineer reviews past payment transactions and submits a refund request for an eligible purchase
Actor	Engineer
Precondition	The engineer is authenticated and has at least one completed payment transaction
Post-condition	Engineer has reviewed their transaction history, or a refund request is created with pending status and submitted for administrative review
Basic Scenario	1. Engineer navigates to the payment history section; system displays all transactions with date, training path, amount, and status; engineer may filter by date range or status 2. Engineer selects a transaction to view its details 3. For Refund: engineer clicks the refund option on an eligible transaction; system displays refund eligibility and a request form; engineer provides a reason and submits; system creates a pending refund request, notifies administrators for review, and allows the engineer to track the request status
Error Scenario	1. If no payment history exists, system displays an empty state 2. If the selected transaction is not eligible for a refund, system displays the refund policy 3. If a refund request already exists for the transaction, system shows its current status

3.2.2.5 Refinement of the Use Case Manage Account

Figure 1.9 presents the refinement of the Use Case Manage Account for the Engineer actor. This use case includes the following sub-use cases:

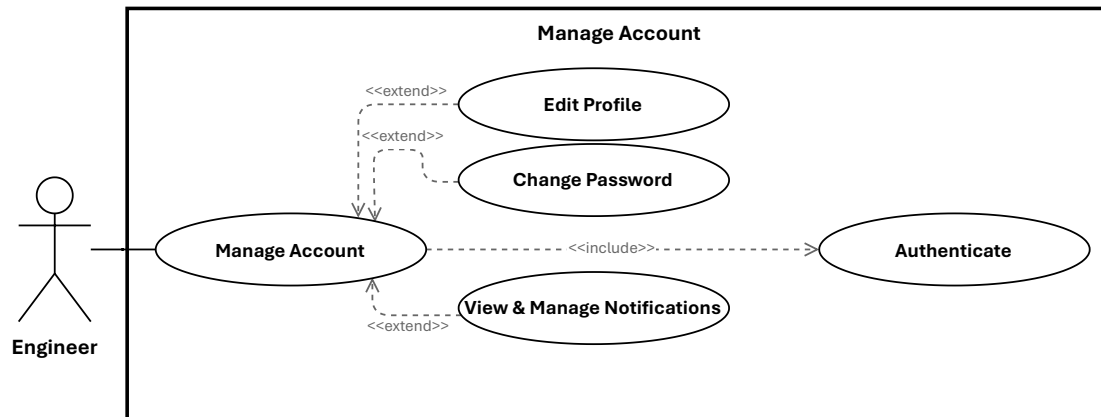


Figure 1.9: Use Case Manage Account

• Specification of the Use Case Edit Profile

Table 1.24: Use Case Specification: Edit Profile

Title	Edit Profile
Summary	Engineer updates their personal profile information including name, bio, and preferences
Actor	Engineer
Precondition	The engineer is authenticated
Post-condition	Profile information is updated and persisted
Basic Scenario	1. Engineer navigates to profile settings 2. System displays current profile information 3. Engineer updates name, bio, or other profile fields 4. System validates the updated information 5. System persists the changes 6. Profile is updated across the platform
Error Scenario	1. If validation fails, the system displays specific error messages 2. If update fails, the system preserves the previous profile data

• Specification of the Use Case Change Password

Table 1.25: Use Case Specification: Change Password

Title	Change Password
Summary	Engineer changes their account password for security purposes
Actor	Engineer
Precondition	The engineer is authenticated and knows their current password
Post-condition	The password is updated and the engineer must authenticate with the new password
Basic Scenario	1. Engineer navigates to password settings 2. System displays password change form 3. Engineer enters current password 4. Engineer enters new password and confirms it 5. System validates current password and new password requirements 6. System updates the password hash 7. System confirms the change and may require re-authentication
Error Scenario	1. If current password is incorrect, the system rejects the change 2. If new password does not meet requirements, the system displays validation errors 3. If passwords do not match, the system prompts for confirmation

- **Specification of the Use Case View & Manage Notifications**

Table 1.26: Use Case Specification: View & Manage Notifications

Title	View & Manage Notifications
Summary	Engineer views system notifications and manages their read/unread status
Actor	Engineer
Precondition	The engineer is authenticated and has received notifications
Post-condition	Engineer views notifications and can mark them as read or take actions
Basic Scenario	1. Engineer navigates to notifications section 2. System displays list of notifications with read/unread status 3. System shows notification type, content, and timestamp 4. Engineer can click a notification to view details 5. Engineer can mark notifications as read or unread 6. Engineer can dismiss or delete notifications
Error Scenario	1. If no notifications exist, the system displays an empty-state message

3.2.3 Teacher Use Cases

The Teacher actor is centered on pedagogical production and publication workflows. The most relevant use cases are training path creation, content maintenance, quiz publication, and submission for approval.

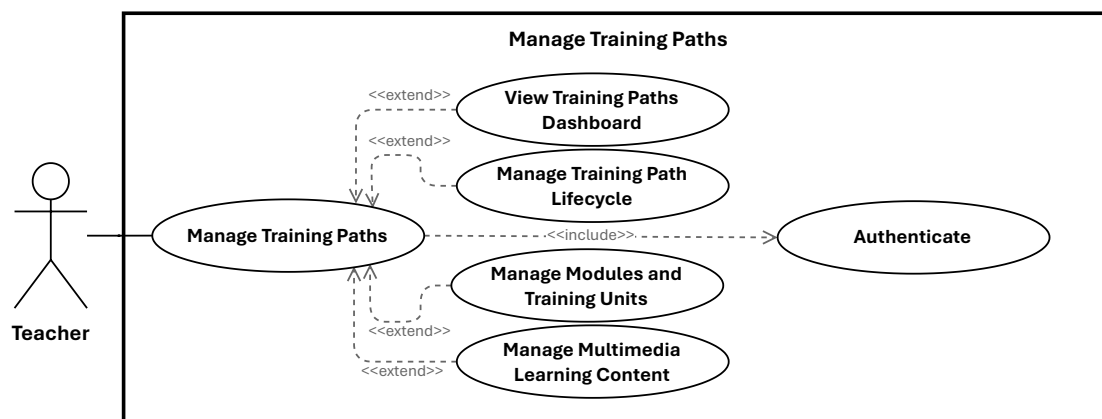
- **Specification of the Use Case View Analytics & Performance**

Table 1.27: Use Case Specification: View Analytics & Performance

Title	View Analytics & Performance
Summary	Teacher reviews enrollment statistics, completion rates, and learner performance for their training paths
Actor	Teacher
Precondition	The teacher is authenticated and has at least one published training path
Post-condition	The teacher views analytics data for their training paths
Basic Scenario	1. Teacher opens the teaching analytics dashboard 2. System retrieves enrollment counts, completion rates, and average quiz scores per training path 3. System displays charts and tables with time-series trends 4. Teacher may filter analytics by date range or training path 5. Teacher may drill down into individual module or unit performance
Error Scenario	1. If no training paths exist, the system displays an empty-state message 2. If analytics data is still computing, the system shows a loading indicator

3.2.3.1 Refinement of the Use Case Manage Training Paths

Figure 1.10 presents the refinement of the Use Case Manage Training Paths for the Teacher actor. This use case includes the following sub-use cases:

**Figure 1.10:** Use Case Manage Training Paths

- **Specification of the Use Case View Training Paths Dashboard**

Table 1.28: Use Case Specification: View Training Paths Dashboard

Title	View Training Paths Dashboard
Summary	Teacher views all owned training paths with statistics including total engineers, average ratings, and completion rates
Actor	Teacher
Precondition	Teacher is authenticated, verified, and approved to teach
Post-condition	Teacher views the list of owned training paths with key metrics
Basic Scenario	1. Teacher navigates to the teaching dashboard 2. System retrieves all training paths owned by the teacher 3. System calculates and displays statistics per path 4. Teacher may click any training path to view or edit details
Error Scenario	1. If no training paths exist, system displays empty state with create option 2. If teacher is not approved, system redirects to pending approval page

- **Specification of the Use Case Manage Training Path Lifecycle**

Table 1.29: Use Case Specification: Manage Training Path Lifecycle

Title	Manage Training Path Lifecycle
Summary	Teacher creates, edits, deletes, archives, restores, and submits training paths for review throughout their lifecycle
Actor	Teacher
Precondition	Teacher is authenticated, verified, approved to teach, and owns the target training path where applicable
Post-condition	The training path is created, updated, removed, archived, restored, or submitted for review according to the action performed
Basic Scenario	1. Teacher selects a lifecycle action from the dashboard: create, edit, delete, archive, restore, or submit for review 2. For Create: Teacher fills in title, description, category, price, and difficulty, then submits; system creates the path with draft status 3. For Edit: Teacher modifies desired metadata fields and submits; system validates and persists changes 4. For Delete: Teacher confirms permanent deletion; system cascades removal of all associated content 5. For Archive: Teacher confirms archiving; system soft-deletes the path and removes it from public listings 6. For Restore: Teacher selects an archived path and confirms restoration; system returns it to its previous state 7. For Submit for Review: Teacher confirms submission; system transitions the path to pending review and notifies administrators
Error Scenario	1. Missing or invalid fields cause validation failure and surface specific error messages 2. Deletion is blocked if active enrollments exist 3. Submission is rejected if the training path is incomplete 4. Unauthorized actions are denied and access is blocked 5. Persistence failures preserve the previous state and surface a recoverable error

• **Specification of the Use Case Manage Modules and Training Units**

Table 1.30: Use Case Specification: Manage Modules and Training Units

Title	Manage Modules and Training Units
Summary	Teacher adds, edits, reorders, and deletes modules and their contained training units to structure the learning content of a training path
Actor	Teacher
Precondition	Teacher is authorized to edit the target training path
Post-condition	The training path structure is updated to reflect the added, modified, reordered, or removed modules and training units
Basic Scenario	1. Teacher opens the training path editor 2. Teacher selects an action on a module or training unit: add, edit, reorder, or delete 3. For Add: Teacher enters title and description and submits; system creates the item and appends it to the structure 4. For Edit: Teacher modifies desired fields and submits; system validates and persists updates 5. For Reorder: Teacher drags items to the desired position; system persists the new order 6. For Delete: Teacher confirms deletion; system removes the item and all contained content with a cascading delete 7. System reflects all structural changes in the editor view
Error Scenario	1. Missing required fields surface validation errors 2. Deletion of items with active engineer progress is warned or blocked 3. Unauthorized access is denied 4. Persistence failures preserve the previous structure and surface a recoverable error

• **Specification of the Use Case Manage Multimedia Learning Content**

Table 1.31: Use Case Specification: Manage Multimedia Learning Content

Title	Manage Multimedia Learning Content
Summary	Teacher creates or updates articles, videos, captions, and quizzes associated with a training unit
Actor	Teacher
Precondition	Teacher is authorized to edit the target training path and training unit
Post-condition	The selected educational asset is created, updated, published, unpublished, or deleted according to the action performed
Basic Scenario	1. Teacher selects a training unit and opens the content editor 2. For Article: Teacher writes or edits rich text content with formatting and media; system persists on save 3. For Video: Teacher uploads a video file; system validates format and size, stores the file, and queues it for transcoding; Teacher may monitor transcoding status, retry on failure, and manage caption files per language 4. For Quiz: Teacher creates a quiz with title, passing score, and time limit; adds, reorders, and manages multiple-choice questions with marked correct answers; then publishes or unpublishes the quiz; Teacher may also view attempt statistics including pass rates and average scores 5. System validates content structure and ownership at each step 6. System persists the asset and updates the training unit status accordingly
Error Scenario	1. Invalid or incomplete content causes validation failure with specific messages 2. Unauthorized editing is blocked 3. Video format or size violations prevent upload 4. Media processing failure leaves the asset in a retryable state 5. Quiz publication is blocked if no questions exist

3.2.3.2 Refinement of the Use Case Manage Forum Interactions

Figure 1.11 presents the refinement of the Use Case Manage Forum Interactions for the Teacher actor. This use case includes the following sub-use cases:

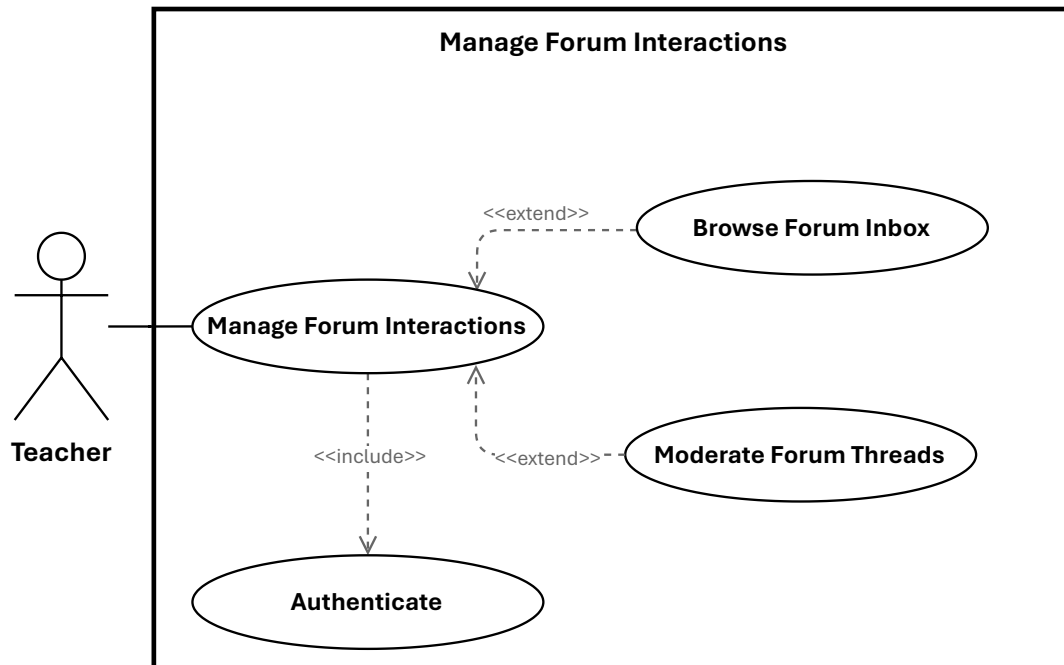


Figure 1.11: Use Case Manage Forum Interactions

- **Specification of the Use Case Browse Forum Inbox**

Table 1.32: Use Case Specification: Browse Forum Inbox

Title	Browse Forum Inbox
Summary	Teacher opens the discussion forum inbox and filters threads by status to identify questions requiring attention
Actor	Teacher
Precondition	Teacher owns at least one training path with forum activity
Post-condition	Teacher views a filtered list of threads and may proceed to read or respond
Basic Scenario	1. Teacher navigates to the forum inbox from the teaching dashboard 2. System displays threads from all owned training paths with reply counts and timestamps 3. Teacher selects a filter to narrow the view: • Flagged — shows threads or replies that have been reported and require moderation • Unanswered — shows threads with no teacher response yet • Recent — shows threads sorted by most recent activity 4. System retrieves and displays threads matching the selected filter 5. Teacher may select any thread to read, respond, or take a moderation action
Error Scenario	1. If no threads match the active filter, system displays an empty state 2. If filter retrieval fails, system falls back to showing all threads with an error notice 3. If teacher has no training paths, system displays an empty state with setup instructions

• **Specification of the Use Case Moderate Forum Threads**

Table 1.33: Use Case Specification: Moderate Forum Threads

Title	Moderate Forum Threads
Summary	Teacher moderates discussion threads within owned training paths by pinning, locking, resolving flags, and editing or removing inappropriate content
Actor	Teacher
Precondition	Teacher owns the training path containing the target thread or reply
Post-condition	Thread state is updated according to the moderation action performed; a log entry is created and affected engineers are notified where applicable
Basic Scenario	1. Teacher opens a thread from the forum inbox 2. Teacher selects a moderation action: • Pin / Unpin — promotes the thread to the top of the forum listing or returns it to chronological order • Lock / Unlock — prevents new replies from engineers or reopens the thread for discussion • Resolve Flag — teacher reviews flagged content, takes appropriate action (approve, edit, or delete), then marks the flag as resolved • Edit / Delete post — teacher corrects or removes inappropriate content and system logs the action with timestamp and reason 3. System validates ownership and applies the requested state change 4. System confirms the action and notifies affected engineers where applicable
Error Scenario	1. Unauthorized access is denied if teacher does not own the training path 2. If the target thread or post has already been removed, system displays a notice 3. Resolving a flag on non-flagged content surfaces an explanatory error 4. Persistence failures preserve the previous state and surface a recoverable error

3.2.3.3 Refinement of the Use Case Manage VM Assignments

Figure ?? presents the refinement of the Use Case Manage VM Assignments for the Teacher actor. This use case includes the following sub-use cases: • **Specification of the Use Case**

Submit VM Assignment

Table 1.34: Use Case Specification: Submit VM Assignment

Title	Submit VM Assignment
Summary	Teacher creates a VM assignment request for a training unit, specifying template, duration, and justification
Actor	Teacher
Precondition	Teacher is authorized to request VMs for the target training unit
Post-condition	A new VM assignment record is created with pending status and notifications are dispatched as configured
Basic Scenario	1. Teacher opens the VM assignment form for a training unit 2. Teacher selects a VM template and duration, and provides a justification 3. System validates ownership and field values, then creates the assignment record 4. System confirms submission and displays the new assignment with its pending status
Error Scenario	1. Validation failures surface field-level errors and preserve input 2. If quota or node availability prevents assignment, system displays an explanatory error

• **Specification of the Use Case Manage Assignments**

Table 1.35: Use Case Specification: Manage VM Assignments

Title	Manage VM Assignments
Summary	Teacher views and cancels VM assignment requests associated with their training paths
Actor	Teacher
Precondition	Teacher is authenticated and associated with at least one training path
Post-condition	Teacher views current assignments or a selected assignment is cancelled and any scheduled provisioning is aborted
Basic Scenario	1. Teacher opens the VM assignments view 2. System displays assignments grouped by training path with status, template, and timestamps 3. Teacher may filter or sort the list, or open an assignment for details 4. To cancel, teacher selects an assignment and confirms the delete action 5. System verifies the assignment is in a cancellable state, removes or marks it cancelled, and updates the list
Error Scenario	1. If no assignments exist, system displays an empty state with setup instructions 2. If the assignment is already provisioned or in a non-cancellable state, system displays an explanatory error 3. Persistence failures preserve current state and surface a recoverable error

- **Specification of the Use Case Manage Earnings & Payouts**

Table 1.36: Use Case Specification: Manage Earnings & Payouts

Title	Manage Earnings & Payouts
Summary	Teacher views their earnings summary, requests a payout of available balance, and exports transaction history as a CSV report
Actor	Teacher
Precondition	Teacher is authenticated and has at least one training path with paid enrollments
Post-condition	Teacher has viewed financial data, submitted a payout request, or downloaded an earnings report according to the action performed
Basic Scenario	1. Teacher navigates to the earnings section 2. System displays total earnings, available balance, pending payouts, and a revenue breakdown by training path; teacher may filter by date range 3. For Request Payout: teacher enters the desired amount and selects a payment method; system validates the amount against the available balance and minimum threshold, creates a pending payout request, and confirms submission 4. For Export Report: teacher specifies a date range and clicks export; system generates a CSV file of transaction details and initiates the download
Error Scenario	1. If no earnings data exists, system displays an empty state 2. Payout amount exceeding available balance or falling below the minimum threshold surfaces a specific validation error 3. Invalid payment method prevents payout submission 4. An invalid or empty date range prevents CSV export 5. Service or export failures surface a recoverable error with a retry option

3.2.4 Administrator Use Cases

The Administrator actor coordinates governance, operations, and infrastructure. The administrator's responsibilities span nine major functional areas: dashboard and platform analytics, user and role management, infrastructure and hardware management, training

path management, reservation management, VM assignment approval, payout and refund processing, maintenance mode configuration, and forum moderation.

3.2.4.1 Refinement of the Use Case Consult Dashboard & Platform Analytics

Figure 1.20 presents the refinement of the Use Case Consult Dashboard & Platform Analytics for the Administrator actor. This use case includes the following sub-use cases:

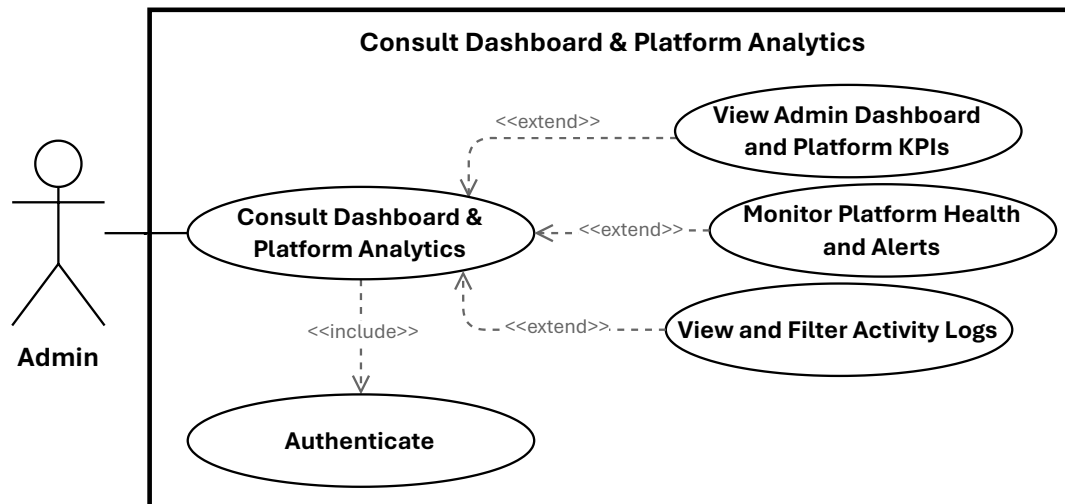


Figure 1.12: Use Case Consult Dashboard & Platform Analytics

- **Specification of the Use Case View Admin Dashboard and Platform KPIs**

Table 1.37: Use Case Specification: View Admin Dashboard and Platform KPIs

Title	View Admin Dashboard and Platform KPIs
Summary	Administrator views system-wide key performance indicators including user statistics, revenue, active sessions, and infrastructure health, with the ability to drill down into detailed breakdowns per metric category
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator has viewed current system KPIs, trend charts, and infrastructure status at the desired level of detail
Basic Scenario	1. Administrator opens the admin dashboard; system aggregates and displays real-time KPI cards covering active VM sessions, registered users, pending approvals, revenue, and infrastructure status indicators 2. Administrator may filter the dashboard by date range or metric category 3. Administrator drills down into a KPI card to access the detailed platform metrics view, which displays time-series charts and comparative breakdowns across user statistics (total, active, new registrations), revenue data (total, monthly, payout amounts), and session data (active sessions, duration, resource utilization) 4. Administrator may further filter detailed metrics by date range, user role, or revenue category
Error Scenario	1. If the metrics or KPI service is unavailable, system displays cached data with a staleness warning 2. If no data exists for the selected period, system shows an empty state message

• **Specification of the Use Case Monitor Platform Health and Alerts**

Table 1.38: Use Case Specification: Monitor Platform Health and Alerts

Title	Monitor Platform Health and Alerts
Summary	Administrator reviews the overall health status of platform services, inspects active system alerts with their severity and context, and acknowledges or resolves alerts to track response actions
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator has reviewed service health and alert status; any addressed alerts are updated to acknowledged or resolved and recorded accordingly
Basic Scenario	1. Administrator navigates to the platform health section; system displays uptime, error rates, and response times for each service (database, cache, queue, external APIs), highlighting any critical or warning conditions 2. Administrator may drill down into individual service health details and configure health check thresholds and alert rules 3. Administrator opens the system alerts panel; system displays a prioritized list of active alerts with severity levels; administrator selects an alert to view detailed context including timestamp, severity, affected resources, and suggested actions; administrator may filter by severity, type, or time range 4. For Acknowledge / Resolve: administrator selects an active alert, enters resolution notes or action taken, and confirms; system updates the alert status, records the resolution, and removes the alert from the active list; administrator may configure automatic alert resolution rules
Error Scenario	1. If the health monitoring service is unavailable, system displays a connection error with a retry option 2. If a service is unreachable, system marks it as critical and suggests investigation 3. If the alerting service is unreachable, system displays a connection error with a retry option 4. If an alert references a deleted resource, system marks it as stale and suggests dismissal 5. If an alert has already been resolved, system displays a notice 6. If a resolution update fails, system preserves the previous state and logs the error

• Specification of the Use Case View and Filter Activity Logs

Table 1.39: Use Case Specification: View and Filter Activity Logs

Title	View and Filter Activity Logs
Summary	Administrator views a summary of recent platform activity and browses the full paginated activity log, applying filters by date range, user, event type, or severity to locate specific events, with the option to export results for compliance or auditing
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges and the activity logging system is active
Post-condition	The administrator has reviewed recent or filtered activity log entries and optionally exported the results
Basic Scenario	1. Administrator navigates to the activity section of the admin dashboard; system displays a summary of recent activity grouped by type (user actions, content updates, system events) with timestamps; administrator may filter by activity type or time range and click any entry for detailed information 2. Administrator navigates to the full Activity Logs view; system retrieves and displays the most recent log entries with pagination, showing timestamp, user, action, and affected resource per entry 3. Administrator applies filters (date range, user, event type, severity); system updates the display to match the criteria; administrator may save filter configurations for future use 4. Administrator exports filtered log entries to CSV or PDF for compliance or auditing purposes
Error Scenario	1. If no entries match the filter criteria, system displays an empty state with a descriptive message 2. If the filter query is too complex, system suggests simplifying the criteria 3. If log storage or the activity service is temporarily unavailable, system displays an error notification with a retry option

3.2.4.2 Refinement of the Use Case Manage Users & Roles

Figure 1.20 presents the refinement of the Use Case Manage Users & Roles for the Administrator actor. This use case includes the following sub-use cases:

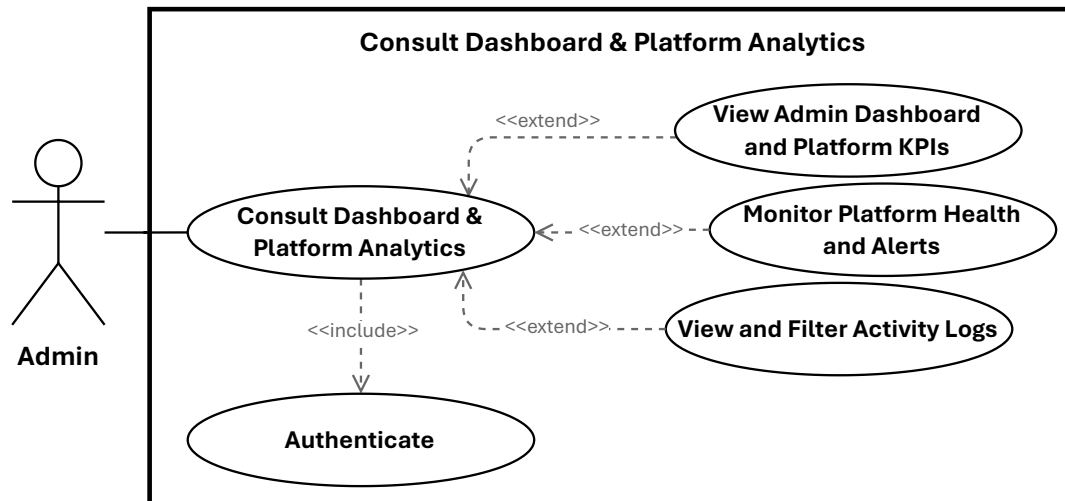


Figure 1.13: Use Case Manage Users & Roles

- **List All Users**

Table 1.40: Use Case Specification: List All Users

Title	List All Users
Summary	Administrator views a paginated list of all registered users with their roles, status, and registration dates
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of users with filtering and sorting options
Basic Scenario	1. Administrator navigates to the User Management section 2. System retrieves all registered users with their profiles 3. System displays users with name, email, role, status, and registration date 4. Administrator may filter users by role, status, or search by name/email 5. Administrator may sort users by registration date, name, or activity 6. Administrator can click on a user to view detailed information
Error Scenario	1. If no users exist, the system displays an empty-state message 2. If the user service is unavailable, the system displays an error notification

- **Specification of the Use Case View User Details**

Table 1.41: Use Case Specification: View User Details

Title	View User Details
Summary	Administrator views detailed information about a specific user including profile, activity history, and account status
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges and has selected a user to view
Post-condition	The administrator views comprehensive user details and can perform account management actions
Basic Scenario	1. Administrator selects a user from the user list 2. System retrieves detailed user information: profile, role, status, activity history 3. System displays user details with sections: profile, account, activity, enrollments 4. Administrator can view user's training path enrollments and progress 5. Administrator can view user's payment history and refund requests 6. Administrator can perform account actions: suspend, unsuspend, change role
Error Scenario	1. If the user does not exist, the system displays a not-found error 2. If the user service is unavailable, the system displays an error notification

- **Specification of the Use Case Approve Teacher Account**

Table 1.42: Use Case Specification: Approve Teacher Account

Title	Approve Teacher Account
Summary	Administrator reviews and approves teacher registration applications before they can publish content
Actor	Administrator
Precondition	The administrator is authenticated; at least one teacher account is pending approval
Post-condition	The teacher account status is updated to approved and the teacher can publish content
Basic Scenario	1. Administrator opens the pending teacher accounts list 2. System displays each applicant's profile, credentials, and submitted documents 3. Administrator reviews the teacher's qualifications and verification documents 4. Administrator approves the account, granting teacher-level privileges 5. System sends a notification to the teacher confirming their approval 6. Teacher can now create and submit training paths for review
Error Scenario	1. If the teacher's documents are incomplete, the administrator requests additional information 2. If the approval service fails, the system displays an error and retains the pending state

- **Specification of the Use Case Revoke Teacher Approval**

Table 1.43: Use Case Specification: Revoke Teacher Approval

Title	Revoke Teacher Approval
Summary	Administrator revokes teacher privileges from an approved teacher account, preventing further content publication
Actor	Administrator
Precondition	The administrator is authenticated and the target user has teacher privileges
Post-condition	The teacher account status is updated to non-teacher and existing published content remains accessible
Basic Scenario	1. Administrator navigates to the user details page for a teacher 2. Administrator selects the revoke teacher approval action 3. System confirms the revocation action and requires a reason 4. Administrator provides the revocation reason and confirms 5. System updates the user's role to non-teacher 6. System sends a notification to the user about the revocation 7. User can no longer create or submit new training paths
Error Scenario	1. If the user is not a teacher, the system displays an error 2. If the revocation fails, the system displays an error and retains the current state

- **Specification of the Use Case Suspend User Account**

Table 1.44: Use Case Specification: Suspend User Account

Title	Suspend User Account
Summary	Administrator suspends a user account, preventing login and access to platform features
Actor	Administrator
Precondition	The administrator is authenticated and the target user account is active
Post-condition	The user account status is updated to suspended and the user cannot log in or access the platform
Basic Scenario	1. Administrator navigates to the user details page 2. Administrator selects the suspend account action 3. System confirms the suspension action and requires a reason 4. Administrator provides the suspension reason and confirms 5. System updates the user's status to suspended 6. System terminates all active sessions for the user 7. System sends a notification to the user about the suspension
Error Scenario	1. If the user is already suspended, the system displays a notice 2. If the suspension fails, the system displays an error and retains the current state

- **Specification of the Use Case Unsuspend User Account**

Table 1.45: Use Case Specification: Unsuspend User Account

Title	Unsuspend User Account
Summary	Administrator unsuspends a user account, restoring login and access to platform features
Actor	Administrator
Precondition	The administrator is authenticated and the target user account is suspended
Post-condition	The user account status is updated to active and the user can log in and access the platform
Basic Scenario	1. Administrator navigates to the user details page 2. Administrator selects the unsuspend account action 3. System confirms the unsuspension action 4. Administrator confirms the unsuspension 5. System updates the user's status to active 6. System sends a notification to the user about the account restoration 7. User can now log in and access the platform
Error Scenario	1. If the user is not suspended, the system displays a notice 2. If the unsuspension fails, the system displays an error and retains the current state

- **Specification of the Use Case Change User Role**

Table 1.46: Use Case Specification: Change User Role

Title	Change User Role
Summary	Administrator changes a user's role to engineer, teacher, or admin, updating their permissions and access
Actor	Administrator
Precondition	The administrator is authenticated and the target user account exists
Post-condition	The user's role is updated and their permissions are changed accordingly
Basic Scenario	1. Administrator navigates to the user details page 2. Administrator selects the change role action 3. System displays available roles: engineer, teacher, admin 4. Administrator selects the new role and confirms 5. System validates the role change (e.g., cannot remove last admin) 6. System updates the user's role 7. System sends a notification to the user about the role change
Error Scenario	1. If the role change would leave no admins, the system prevents the action 2. If the role change fails, the system displays an error and retains the current state

- **Specification of the Use Case Impersonate User (Debug)**

Table 1.47: Use Case Specification: Impersonate User (Debug)

Title	Impersonate User (Debug)
Summary	Administrator impersonates a user account to debug issues or view the platform from the user's perspective
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges and the target user account exists
Post-condition	The administrator is logged in as the target user and can perform actions as that user
Basic Scenario	1. Administrator navigates to the user details page 2. Administrator selects the impersonate user action 3. System confirms the impersonation action 4. Administrator confirms the impersonation 5. System creates an impersonation session for the administrator 6. Administrator is logged in as the target user 7. System logs the impersonation action for audit purposes
Error Scenario	1. If the user does not exist, the system displays an error 2. If the impersonation fails, the system displays an error and retains the current session

- **Specification of the Use Case GDPR User Data Deletion**

Table 1.48: Use Case Specification: GDPR User Data Deletion

Title	GDPR User Data Deletion
Summary	Administrator permanently deletes a user account and all associated data in compliance with GDPR requirements
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges and the target user account exists
Post-condition	The user account and all associated data are permanently deleted from the system
Basic Scenario	1. Administrator navigates to the user details page 2. Administrator selects the GDPR data deletion action 3. System displays a warning about permanent data deletion 4. System lists all data that will be deleted: profile, enrollments, payments, activity logs 5. Administrator confirms the deletion action 6. System permanently deletes the user account and all associated data 7. System logs the deletion action for audit purposes
Error Scenario	1. If the user does not exist, the system displays an error 2. If the deletion fails, the system displays an error and retains the current state

3.2.4.3 Refinement of the Use Case Manage Infrastructure and Hardware

Figure 1.20 presents the refinement of the Use Case Manage Infrastructure and Hardware for the Administrator actor. This use case includes the following sub-use cases:

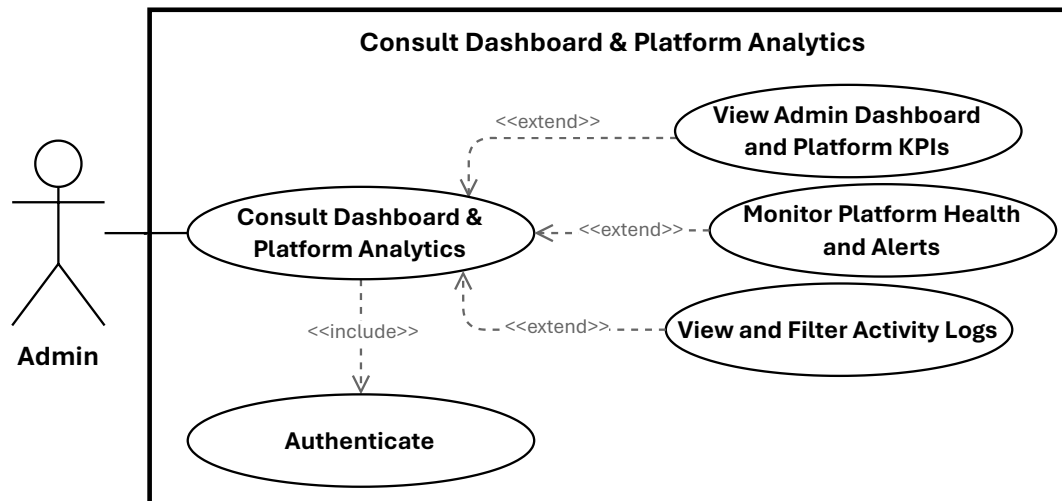


Figure 1.14: Use Case Manage Infrastructure and Hardware

• Specification of the Use Case List Proxmox Servers

Table 1.49: Use Case Specification: List Proxmox Servers

Title	List Proxmox Servers
Summary	Administrator views all registered Proxmox servers with their connection status and node information
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of Proxmox servers with status and node counts
Basic Scenario	1. Administrator navigates to the Proxmox Servers section 2. System retrieves all registered Proxmox servers 3. System displays servers with name, host, status, and node count 4. Administrator can view server details and connection status 5. Administrator can perform server management actions
Error Scenario	1. If no servers are registered, the system displays an empty-state message 2. If the server list cannot be retrieved, the system shows an error notification

• Specification of the Use Case Test Server Connection

Table 1.50: Use Case Specification: Test Server Connection

Title	Test Server Connection
Summary	Administrator tests the connection to a Proxmox server to verify credentials and network accessibility
Actor	Administrator
Precondition	The administrator is authenticated and has selected a Proxmox server to test
Post-condition	The connection test result is displayed with success or failure details
Basic Scenario	1. Administrator selects a Proxmox server from the server list 2. Administrator clicks the Test Connection button 3. System attempts to connect to the Proxmox API using stored credentials 4. System displays the connection result: success, timeout, or authentication error 5. If successful, system shows server version and node information
Error Scenario	1. If the connection times out, the system displays a timeout error 2. If authentication fails, the system displays an authentication error 3. If the server is unreachable, the system displays a network error

- **Specification of the Use Case Register Proxmox Server**

Table 1.51: Use Case Specification: Register Proxmox Server

Title	Register Proxmox Server
Summary	Administrator adds a new Proxmox server to the platform infrastructure for VM provisioning
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The Proxmox server is registered and available for node synchronization and VM provisioning
Basic Scenario	1. Administrator navigates to the Proxmox Servers section 2. Administrator clicks Add Server 3. System displays the server registration form 4. Administrator enters server name, host, port, username, and password 5. System validates the connection to the Proxmox API 6. System stores the server credentials securely 7. System adds the server to the infrastructure list
Error Scenario	1. If the connection test fails, the system displays an error and prevents registration 2. If the server name already exists, the system prompts for a unique name 3. If credentials are invalid, the system displays an authentication error

- **Specification of the Use Case Edit Server Credentials**

Table 1.52: Use Case Specification: Edit Server Credentials

Title	Edit Server Credentials
Summary	Administrator updates the connection credentials for a registered Proxmox server
Actor	Administrator
Precondition	The administrator is authenticated and has selected a Proxmox server to edit
Post-condition	The server credentials are updated and the connection is verified
Basic Scenario	1. Administrator selects a Proxmox server from the server list 2. Administrator clicks Edit 3. System displays the server edit form with current credentials 4. Administrator updates the username, password, or connection details 5. System validates the new credentials by testing the connection 6. System stores the updated credentials securely 7. System confirms the update and displays the new connection status
Error Scenario	1. If the new credentials fail authentication, the system displays an error 2. If the connection test fails, the system prevents the update

- **Specification of the Use Case Inactivate Server**

Table 1.53: Use Case Specification: Inactivate Server

Title	Inactivate Server
Summary	Administrator deactivates a Proxmox server to prevent new VM provisioning while preserving existing sessions
Actor	Administrator
Precondition	The administrator is authenticated and has selected an active Proxmox server
Post-condition	The server is marked as inactive and no new VMs are provisioned on it
Basic Scenario	1. Administrator selects a Proxmox server from the server list 2. Administrator clicks Inactivate 3. System confirms the inactivation action 4. System updates the server status to inactive 5. System prevents new VM provisioning on this server 6. Existing VM sessions continue to run normally
Error Scenario	1. If the server is already inactive, the system displays a notice 2. If the inactivation fails, the system displays an error and retains the current state

- **Specification of the Use Case Delete Proxmox Server**

Table 1.54: Use Case Specification: Delete Proxmox Server

Title	Delete Proxmox Server
Summary	Administrator permanently removes a Proxmox server from the platform infrastructure
Actor	Administrator
Precondition	The administrator is authenticated and has selected a Proxmox server with no active VMs
Post-condition	The Proxmox server is removed from the infrastructure and all associated data is deleted
Basic Scenario	1. Administrator selects a Proxmox server from the server list 2. Administrator clicks Delete 3. System displays a warning about permanent deletion 4. System checks for active VMs on the server 5. If no active VMs exist, administrator confirms the deletion 6. System removes the server and all associated data 7. System logs the deletion for audit purposes
Error Scenario	1. If active VMs exist on the server, the system prevents deletion 2. If the deletion fails, the system displays an error and retains the server

- **Specification of the Use Case Sync Nodes from Server**

Table 1.55: Use Case Specification: Sync Nodes from Server

Title	Sync Nodes from Server
Summary	Administrator synchronizes the node list from a Proxmox server to update the platform's infrastructure view
Actor	Administrator
Precondition	The administrator is authenticated and has selected an active Proxmox server
Post-condition	The node list is updated with the latest information from the Proxmox server
Basic Scenario	1. Administrator selects a Proxmox server from the server list 2. Administrator clicks Sync Nodes 3. System connects to the Proxmox API 4. System retrieves the current node list from the server 5. System updates the local node records with new or changed nodes 6. System removes nodes that no longer exist on the server 7. System displays the sync result with added, updated, and removed nodes
Error Scenario	1. If the connection to the Proxmox API fails, the system displays an error 2. If the sync fails, the system preserves the previous node list

- **Specification of the Use Case List Nodes**

Table 1.56: Use Case Specification: List Nodes

Title	List Nodes
Summary	Administrator views all nodes from Proxmox servers with their status and resource utilization
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of nodes with status, CPU, memory, and storage information
Basic Scenario	1. Administrator navigates to the Nodes section 2. System retrieves all nodes from registered Proxmox servers 3. System displays nodes with name, server, status, and resource metrics 4. Administrator can filter nodes by server or status 5. Administrator can view detailed node information
Error Scenario	1. If no nodes are available, the system displays an empty-state message 2. If the node list cannot be retrieved, the system shows an error notification

- **Specification of the Use Case List VMs on Node**

Table 1.57: Use Case Specification: List VMs on Node

Title	List VMs on Node
Summary	Administrator views all virtual machines on a specific Proxmox node with their status and configuration
Actor	Administrator
Precondition	The administrator is authenticated and has selected a Proxmox node
Post-condition	The administrator views a list of VMs with status, resource allocation, and configuration details
Basic Scenario	1. Administrator selects a Proxmox node from the node list 2. System retrieves all VMs from the selected node 3. System displays VMs with name, status, CPU, memory, and disk allocation 4. Administrator can filter VMs by status or resource type 5. Administrator can view detailed VM configuration
Error Scenario	1. If no VMs exist on the node, the system displays an empty-state message 2. If the VM list cannot be retrieved, the system shows an error notification

- **Specification of the Use Case Start VM**

Table 1.58: Use Case Specification: Start VM

Title	Start VM
Summary	Administrator starts a stopped virtual machine on a Proxmox node
Actor	Administrator
Precondition	The administrator is authenticated and has selected a stopped VM on a Proxmox node
Post-condition	The VM is started and becomes available for use
Basic Scenario	1. Administrator selects a stopped VM from the VM list 2. Administrator clicks Start 3. System confirms the start action 4. System sends a start command to the Proxmox API 5. Proxmox starts the VM and updates its status 6. System displays the updated VM status as running
Error Scenario	1. If the VM is already running, the system displays a notice 2. If the start command fails, the system displays an error and retains the stopped state

- **Specification of the Use Case Stop VM**

Table 1.59: Use Case Specification: Stop VM

Title	Stop VM
Summary	Administrator stops a running virtual machine on a Proxmox node
Actor	Administrator
Precondition	The administrator is authenticated and has selected a running VM on a Proxmox node
Post-condition	The VM is stopped and no longer consuming resources
Basic Scenario	1. Administrator selects a running VM from the VM list 2. Administrator clicks Stop 3. System confirms the stop action 4. System sends a stop command to the Proxmox API 5. Proxmox stops the VM and updates its status 6. System displays the updated VM status as stopped
Error Scenario	1. If the VM is already stopped, the system displays a notice 2. If the stop command fails, the system displays an error and retains the running state

- **Specification of the Use Case Reboot VM**

Table 1.60: Use Case Specification: Reboot VM

Title	Reboot VM
Summary	Administrator reboots a running virtual machine on a Proxmox node
Actor	Administrator
Precondition	The administrator is authenticated and has selected a running VM on a Proxmox node
Post-condition	The VM is rebooted and resumes operation with a fresh system state
Basic Scenario	1. Administrator selects a running VM from the VM list 2. Administrator clicks Reboot 3. System confirms the reboot action 4. System sends a reboot command to the Proxmox API 5. Proxmox reboots the VM and updates its status 6. System displays the updated VM status as running after reboot
Error Scenario	1. If the VM is not running, the system displays an error 2. If the reboot command fails, the system displays an error and retains the current state

- **Specification of the Use Case Add Gateway Node**

Table 1.61: Use Case Specification: Add Gateway Node

Title	Add Gateway Node
Summary	Administrator registers a new gateway node for hardware device management and USB forwarding
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The gateway node is registered and available for device discovery and management
Basic Scenario	1. Administrator navigates to the Gateway Nodes section 2. Administrator clicks Add Gateway 3. System displays the gateway registration form 4. Administrator enters gateway name, host, port, and authentication details 5. System validates the connection to the gateway 6. System stores the gateway configuration 7. System adds the gateway to the infrastructure list
Error Scenario	1. If the connection test fails, the system displays an error and prevents registration 2. If the gateway name already exists, the system prompts for a unique name

- **Specification of the Use Case Edit Gateway Node**

Table 1.62: Use Case Specification: Edit Gateway Node

Title	Edit Gateway Node
Summary	Administrator updates the configuration of a registered gateway node
Actor	Administrator
Precondition	The administrator is authenticated and has selected a gateway node to edit
Post-condition	The gateway node configuration is updated and the connection is verified
Basic Scenario	1. Administrator selects a gateway node from the gateway list 2. Administrator clicks Edit 3. System displays the gateway edit form with current configuration 4. Administrator updates the name, host, port, or authentication details 5. System validates the new configuration by testing the connection 6. System stores the updated configuration 7. System confirms the update and displays the new connection status
Error Scenario	1. If the new configuration fails validation, the system displays an error 2. If the connection test fails, the system prevents the update

- **Specification of the Use Case Verify Gateway Node**

Table 1.63: Use Case Specification: Verify Gateway Node

Title	Verify Gateway Node
Summary	Administrator verifies the connectivity and status of a gateway node
Actor	Administrator
Precondition	The administrator is authenticated and has selected a gateway node to verify
Post-condition	The gateway node status is displayed with connectivity and device information
Basic Scenario	1. Administrator selects a gateway node from the gateway list 2. Administrator clicks Verify 3. System connects to the gateway node 4. System retrieves the gateway status and connected device count 5. System displays the verification result with status and device information
Error Scenario	1. If the gateway is unreachable, the system displays a connection error 2. If the verification fails, the system shows an error with diagnostic information

- **Specification of the Use Case Delete Gateway Node**

Table 1.64: Use Case Specification: Delete Gateway Node

Title	Delete Gateway Node
Summary	Administrator removes a gateway node from the platform infrastructure
Actor	Administrator
Precondition	The administrator is authenticated and has selected a gateway node with no active devices
Post-condition	The gateway node is removed from the infrastructure and all associated data is deleted
Basic Scenario	1. Administrator selects a gateway node from the gateway list 2. Administrator clicks Delete 3. System displays a warning about permanent deletion 4. System checks for active devices on the gateway 5. If no active devices exist, administrator confirms the deletion 6. System removes the gateway and all associated data 7. System logs the deletion for audit purposes
Error Scenario	1. If active devices exist on the gateway, the system prevents deletion 2. If the deletion fails, the system displays an error and retains the gateway

- **Specification of the Use Case Discover Hardware Devices**

Table 1.65: Use Case Specification: Discover Hardware Devices

Title	Discover Hardware Devices
Summary	Administrator scans gateway nodes to discover and register available hardware devices
Actor	Administrator
Precondition	The administrator is authenticated and has at least one active gateway node
Post-condition	Available hardware devices are discovered and registered in the platform inventory
Basic Scenario	1. Administrator navigates to the Hardware Devices section 2. Administrator clicks Discover Devices 3. System queries all active gateway nodes for connected devices 4. System retrieves device information: type, model, serial number, and status 5. System registers new devices and updates existing device information 6. System displays the discovery result with added and updated devices
Error Scenario	1. If no gateway nodes are available, the system displays an error 2. If the discovery fails, the system shows an error with diagnostic information

- **Specification of the Use Case Check Hardware Status**

Table 1.66: Use Case Specification: Check Hardware Status

Title	Check Hardware Status
Summary	Administrator views the current status of hardware devices including availability and connection state
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views the status of all hardware devices with their current state
Basic Scenario	1. Administrator navigates to the Hardware Devices section 2. System retrieves the status of all registered hardware devices 3. System displays devices with type, model, status, and gateway information 4. Administrator can filter devices by type, status, or gateway 5. Administrator can view detailed device information
Error Scenario	1. If no devices are registered, the system displays an empty-state message 2. If the device status cannot be retrieved, the system shows an error notification

- **Specification of the Use Case View Running VMs**

Table 1.67: Use Case Specification: View Running VMs

Title	View Running VMs
Summary	Administrator views all currently running virtual machines across all Proxmox nodes
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of running VMs with resource utilization and session information
Basic Scenario	1. Administrator navigates to the Running VMs section 2. System retrieves all VMs with running status from Proxmox nodes 3. System displays VMs with name, node, owner, CPU, memory, and uptime 4. Administrator can filter VMs by node or owner 5. Administrator can view detailed VM information and session details
Error Scenario	1. If no VMs are running, the system displays an empty-state message 2. If the VM list cannot be retrieved, the system shows an error notification

- **Specification of the Use Case Manage Dedicated Devices**

Table 1.68: Use Case Specification: Manage Dedicated Devices

Title	Manage Dedicated Devices
Summary	Administrator configures hardware devices as dedicated for specific users or training paths
Actor	Administrator
Precondition	The administrator is authenticated and has selected a hardware device to configure
Post-condition	The device is marked as dedicated and assigned to the specified user or training path
Basic Scenario	1. Administrator selects a hardware device from the device list 2. Administrator clicks Manage Dedication 3. System displays the dedication configuration form 4. Administrator selects dedication type: user or training path 5. Administrator selects the target user or training path 6. System updates the device dedication settings 7. System confirms the dedication and displays the new configuration
Error Scenario	1. If the device is already in use, the system prevents dedication changes 2. If the dedication update fails, the system displays an error and retains the current state

- **Specification of the Use Case List All Cameras**

Table 1.69: Use Case Specification: List All Cameras

Title	List All Cameras
Summary	Administrator views all registered cameras with their status and configuration
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of cameras with status, PTZ support, and assignment information
Basic Scenario	1. Administrator navigates to the Cameras section 2. System retrieves all registered cameras 3. System displays cameras with name, status, PTZ support, and VM assignment 4. Administrator can filter cameras by status or PTZ capability 5. Administrator can view detailed camera configuration
Error Scenario	1. If no cameras are registered, the system displays an empty-state message 2. If the camera list cannot be retrieved, the system shows an error notification

- **Specification of the Use Case Assign Camera to VM**

Table 1.70: Use Case Specification: Assign Camera to VM

Title	Assign Camera to VM
Summary	Administrator assigns a camera to a virtual machine for remote monitoring access
Actor	Administrator
Precondition	The administrator is authenticated and has selected an available camera and a VM
Post-condition	The camera is assigned to the VM and becomes accessible from within the VM session
Basic Scenario	1. Administrator selects a camera from the camera list 2. Administrator clicks Assign to VM 3. System displays available VMs for assignment 4. Administrator selects the target VM 5. System validates the assignment and updates the camera configuration 6. System confirms the assignment and displays the new configuration
Error Scenario	1. If the camera is already assigned, the system prompts to reassign 2. If the VM is not available, the system displays an error

- **Specification of the Use Case Unassign Camera**

Table 1.71: Use Case Specification: Unassign Camera

Title	Unassign Camera
Summary	Administrator removes a camera assignment from a virtual machine
Actor	Administrator
Precondition	The administrator is authenticated and has selected an assigned camera
Post-condition	The camera is unassigned and becomes available for other VM assignments
Basic Scenario	1. Administrator selects an assigned camera from the camera list 2. Administrator clicks Unassign 3. System confirms the unassignment action 4. System removes the camera assignment 5. System confirms the unassignment and displays the camera as available
Error Scenario	1. If the camera is not assigned, the system displays a notice 2. If the unassignment fails, the system displays an error and retains the assignment

- **Specification of the Use Case Bulk Assign Cameras**

Table 1.72: Use Case Specification: Bulk Assign Cameras

Title	Bulk Assign Cameras
Summary	Administrator assigns multiple cameras to a virtual machine in a single operation
Actor	Administrator
Precondition	The administrator is authenticated and has selected multiple available cameras and a VM
Post-condition	All selected cameras are assigned to the VM and become accessible from within the VM session
Basic Scenario	1. Administrator navigates to the Cameras section 2. Administrator selects multiple cameras from the camera list 3. Administrator clicks Bulk Assign 4. System displays available VMs for assignment 5. Administrator selects the target VM 6. System validates all assignments and updates the camera configurations 7. System confirms the bulk assignment and displays the results
Error Scenario	1. If any camera is already assigned, the system prompts to reassign 2. If the VM is not available, the system displays an error

- **Specification of the Use Case Activate Camera**

Table 1.73: Use Case Specification: Activate Camera

Title	Activate Camera
Summary	Administrator activates a deactivated camera to make it available for use
Actor	Administrator
Precondition	The administrator is authenticated and has selected a deactivated camera
Post-condition	The camera is activated and becomes available for assignment and streaming
Basic Scenario	1. Administrator selects a deactivated camera from the camera list 2. Administrator clicks Activate 3. System confirms the activation action 4. System updates the camera status to active 5. System confirms the activation and displays the camera as available
Error Scenario	1. If the camera is already active, the system displays a notice 2. If the activation fails, the system displays an error and retains the current state

- **Specification of the Use Case Deactivate Camera**

Table 1.74: Use Case Specification: Deactivate Camera

Title	Deactivate Camera
Summary	Administrator deactivates an active camera to prevent its use for streaming or assignment
Actor	Administrator
Precondition	The administrator is authenticated and has selected an active camera
Post-condition	The camera is deactivated and becomes unavailable for assignment and streaming
Basic Scenario	1. Administrator selects an active camera from the camera list 2. Administrator clicks Deactivate 3. System confirms the deactivation action 4. System updates the camera status to inactive 5. System confirms the deactivation and displays the camera as unavailable
Error Scenario	1. If the camera is already inactive, the system displays a notice 2. If the deactivation fails, the system displays an error and retains the current state

- **Specification of the Use Case Assign USB to VM**

Table 1.75: Use Case Specification: Assign USB to VM

Title	Assign USB to VM
Summary	Administrator assigns a USB device to a virtual machine for direct access
Actor	Administrator
Precondition	The administrator is authenticated and has selected an available USB device and a VM
Post-condition	The USB device is assigned to the VM and becomes accessible from within the VM session
Basic Scenario	1. Administrator selects a USB device from the device list 2. Administrator clicks Assign to VM 3. System displays available VMs for assignment 4. Administrator selects the target VM 5. System validates the assignment and updates the device configuration 6. System confirms the assignment and displays the new configuration
Error Scenario	1. If the device is already assigned, the system prompts to reassign 2. If the VM is not available, the system displays an error

- **Specification of the Use Case Unassign USB**

Table 1.76: Use Case Specification: Unassign USB

Title	Unassign USB
Summary	Administrator removes a USB device assignment from a virtual machine
Actor	Administrator
Precondition	The administrator is authenticated and has selected an assigned USB device
Post-condition	The USB device is unassigned and becomes available for other VM assignments
Basic Scenario	1. Administrator selects an assigned USB device from the device list 2. Administrator clicks Unassign 3. System confirms the unassignment action 4. System removes the USB device assignment 5. System confirms the unassignment and displays the device as available
Error Scenario	1. If the device is not assigned, the system displays a notice 2. If the unassignment fails, the system displays an error and retains the assignment

- **Specification of the Use Case Manage Connection Preferences**

Table 1.77: Use Case Specification: Manage Connection Preferences

Title	Manage Connection Preferences
Summary	Administrator configures preferred remote desktop connection settings such as protocol, resolution, and keyboard layout
Actor	Administrator
Precondition	The administrator is authenticated and has at least one VM session or connection profile
Post-condition	The administrator's connection preference profile is created or updated
Basic Scenario	1. Administrator opens the connection preferences settings 2. System displays available protocols (RDP, VNC, SSH) and configurable parameters 3. Administrator selects a default protocol and configures resolution, color depth, and keyboard layout 4. Administrator saves the preference profile 5. System persists the profile and applies it to future Guacamole connection tokens 6. Engineer may create multiple profiles for different session types
Error Scenario	1. If the selected protocol is not supported, the system rejects the configuration 2. If the profile name is a duplicate, the system prompts for a unique name

3.2.4.4 Refinement of the Use Case Set Maintenance Mode

Figure 1.20 presents the refinement of the Use Case Set Maintenance Mode for the Administrator actor. This use case includes the following sub-use cases:

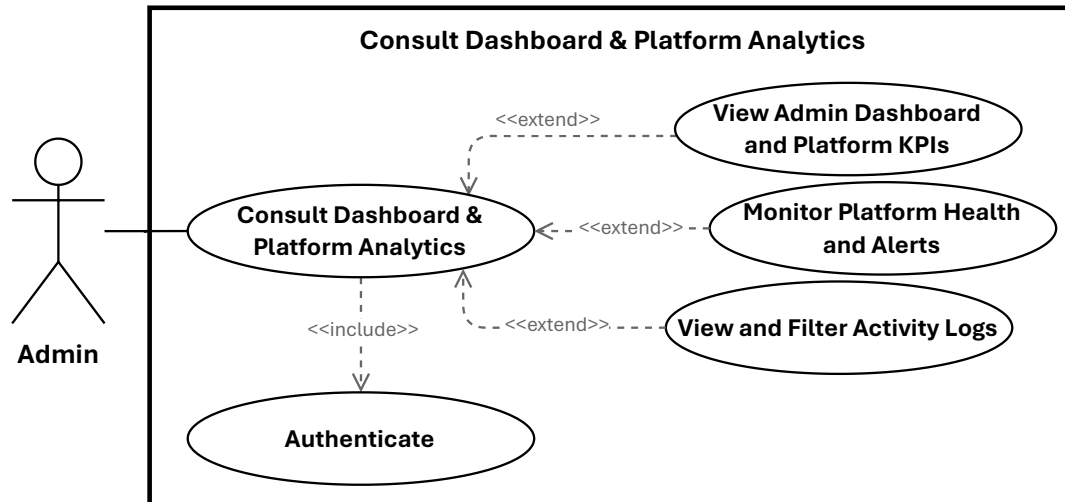


Figure 1.15: Use Case Set Maintenance Mode

• Specification of the Use Case View All Devices in Maintenance

Table 1.78: Use Case Specification: View All Devices in Maintenance

Title	View All Devices in Maintenance
Summary	Administrator views all hardware devices currently in maintenance mode
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of devices in maintenance with their maintenance details
Basic Scenario	1. Administrator navigates to the Maintenance section 2. System retrieves all devices in maintenance mode 3. System displays devices with type, model, maintenance reason, and start date 4. Administrator can filter devices by type or maintenance reason 5. Administrator can view detailed maintenance information
Error Scenario	1. If no devices are in maintenance, the system displays an empty-state message 2. If the device list cannot be retrieved, the system shows an error notification

• Specification of the Use Case Set USB Device Maintenance Description**Table 1.79:** Use Case Specification: Set USB Device Maintenance Description

Title	Set USB Device Maintenance Description
Summary	Administrator adds or updates the maintenance description for a USB device
Actor	Administrator
Precondition	The administrator is authenticated and has selected a USB device in maintenance
Post-condition	The USB device maintenance description is updated and visible to users
Basic Scenario	1. Administrator selects a USB device in maintenance 2. Administrator clicks Edit Maintenance Description 3. System displays the maintenance description form 4. Administrator enters or updates the maintenance description 5. System saves the description and updates the device record 6. System confirms the update and displays the new description
Error Scenario	1. If the device is not in maintenance, the system displays an error 2. If the update fails, the system displays an error and retains the current description

• Specification of the Use Case Enable USB Device Maintenance

Table 1.80: Use Case Specification: Enable USB Device Maintenance

Title	Enable USB Device Maintenance
Summary	Administrator places a USB device into maintenance mode to prevent new reservations
Actor	Administrator
Precondition	The administrator is authenticated and has selected an available USB device
Post-condition	The USB device is placed in maintenance mode and no new reservations can be made
Basic Scenario	1. Administrator selects an available USB device 2. Administrator clicks Enable Maintenance 3. System displays the maintenance form with description field 4. Administrator enters the maintenance reason and description 5. System updates the device status to maintenance 6. System prevents new reservations for the device 7. System confirms the maintenance mode activation
Error Scenario	1. If the device is already in maintenance, the system displays a notice 2. If the maintenance activation fails, the system displays an error and retains the current state

- **Specification of the Use Case Disable USB Device Maintenance**

Table 1.81: Use Case Specification: Disable USB Device Maintenance

Title	Disable USB Device Maintenance
Summary	Administrator removes a USB device from maintenance mode, making it available for reservations
Actor	Administrator
Precondition	The administrator is authenticated and has selected a USB device in maintenance
Post-condition	The USB device is removed from maintenance mode and becomes available for reservations
Basic Scenario	1. Administrator selects a USB device in maintenance 2. Administrator clicks Disable Maintenance 3. System confirms the action 4. System updates the device status to available 5. System clears the maintenance description 6. System confirms the maintenance mode deactivation
Error Scenario	1. If the device is not in maintenance, the system displays a notice 2. If the maintenance deactivation fails, the system displays an error and retains the current state

- **Specification of the Use Case Enable Camera Maintenance Mode**

Table 1.82: Use Case Specification: Enable Camera Maintenance Mode

Title	Enable Camera Maintenance Mode
Summary	Administrator places a camera into maintenance mode to prevent streaming and assignment
Actor	Administrator
Precondition	The administrator is authenticated and has selected an available camera
Post-condition	The camera is placed in maintenance mode and no streaming or assignment can occur
Basic Scenario	1. Administrator selects an available camera 2. Administrator clicks Enable Maintenance 3. System displays the maintenance form with description field 4. Administrator enters the maintenance reason and description 5. System updates the camera status to maintenance 6. System prevents streaming and assignment for the camera 7. System confirms the maintenance mode activation
Error Scenario	1. If the camera is already in maintenance, the system displays a notice 2. If the maintenance activation fails, the system displays an error and retains the current state

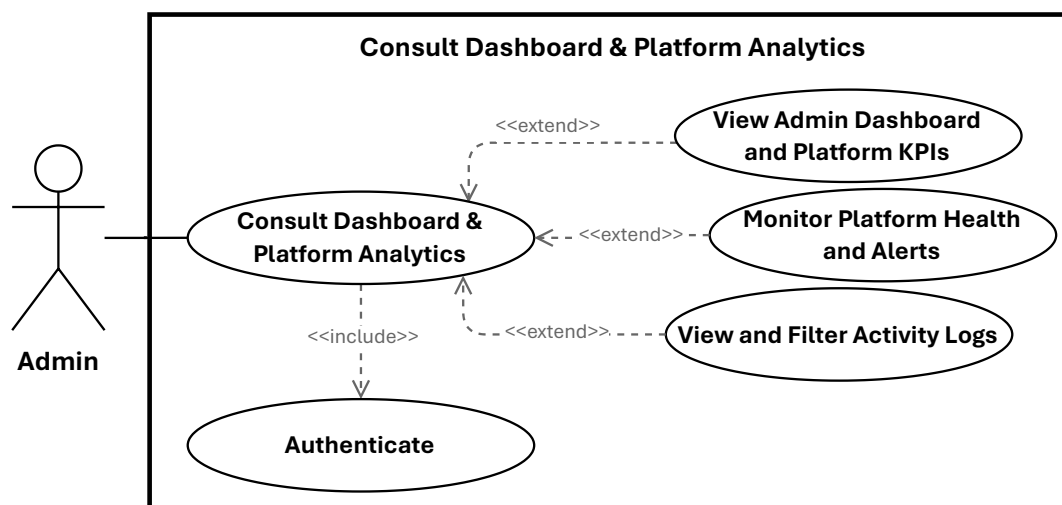
- **Specification of the Use Case Disable Camera Maintenance Mode**

Table 1.83: Use Case Specification: Disable Camera Maintenance Mode

Title	Disable Camera Maintenance Mode
Summary	Administrator removes a camera from maintenance mode, making it available for streaming and assignment
Actor	Administrator
Precondition	The administrator is authenticated and has selected a camera in maintenance
Post-condition	The camera is removed from maintenance mode and becomes available for streaming and assignment
Basic Scenario	1. Administrator selects a camera in maintenance 2. Administrator clicks Disable Maintenance 3. System confirms the action 4. System updates the camera status to available 5. System clears the maintenance description 6. System confirms the maintenance mode deactivation
Error Scenario	1. If the camera is not in maintenance, the system displays a notice 2. If the maintenance deactivation fails, the system displays an error and retains the current state

3.2.4.5 Refinement of the Use Case Moderate Forum Threads

Figure 1.20 presents the refinement of the Use Case Moderate Forum Threads for the Administrator actor. This use case includes the following sub-use cases:

**Figure 1.16:** Use Case Moderate Forum Threads

- **Specification of the Use Case View Flagged Threads & Replies**

Table 1.84: Use Case Specification: View Flagged Threads & Replies

Title	View Flagged Threads & Replies
Summary	Administrator views a list of flagged forum threads and replies that have been reported by users
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges; there are flagged forum posts pending review
Post-condition	The administrator views the list of flagged posts with details and can take moderation actions
Basic Scenario	1. Administrator navigates to the Forum Moderation section 2. System displays a list of flagged forum posts with flag reason, reporter, and post content 3. Administrator can filter flagged posts by severity, type, or time range 4. Administrator can view the full thread context for each flagged post 5. Administrator can select a post to take moderation action
Error Scenario	1. If no flagged posts exist, the system displays an empty-state message 2. If the forum service is unavailable, the system displays an error notification

- **Specification of the Use Case Unflag Thread**

Table 1.85: Use Case Specification: Unflag Thread

Title	Unflag Thread
Summary	Administrator removes the flag from a forum thread, dismissing the report and restoring the thread to normal status
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges and there is a flagged thread to unflag
Post-condition	The thread flag is removed and the thread is no longer in the moderation queue
Basic Scenario	1. Administrator selects a flagged thread from the moderation queue 2. Administrator reviews the thread content and flag reason 3. Administrator selects the unflag action 4. System removes the flag from the thread 5. System removes the thread from the moderation queue 6. System logs the unflag action for audit purposes
Error Scenario	1. If the thread has already been unflagged, the system displays a notice 2. If the unflag action fails, the system displays an error and retains the current state

- **Specification of the Use Case Unflag Reply**

Table 1.86: Use Case Specification: Unflag Reply

Title	Unflag Reply
Summary	Administrator removes the flag from a forum reply, dismissing the report and restoring the reply to normal status
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges and there is a flagged reply to unflag
Post-condition	The reply flag is removed and the reply is no longer in the moderation queue
Basic Scenario	1. Administrator selects a flagged reply from the moderation queue 2. Administrator reviews the reply content and flag reason 3. Administrator selects the unflag action 4. System removes the flag from the reply 5. System removes the reply from the moderation queue 6. System logs the unflag action for audit purposes
Error Scenario	1. If the reply has already been unflagged, the system displays a notice 2. If the unflag action fails, the system displays an error and retains the current state

3.2.4.6 Refinement of the Use Case Manage Training Paths

Figure 1.20 presents the refinement of the Use Case Manage Training Paths for the Administrator actor. This use case includes the following sub-use cases:

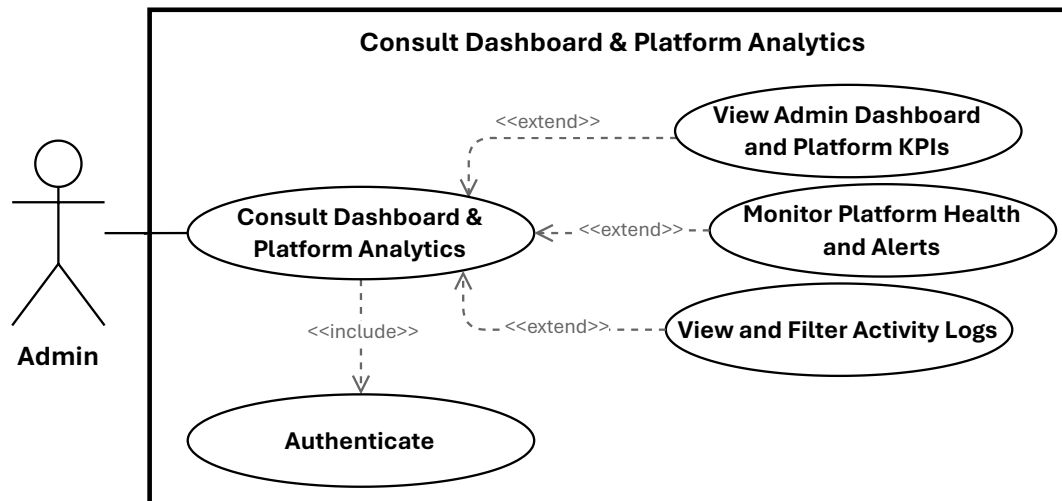


Figure 1.17: Use Case Manage Training Paths

• Specification of the Use Case List All Training Paths

Table 1.87: Use Case Specification: List All Training Paths

Title	List All Training Paths
Summary	Administrator views all training paths with their status, author, and approval state
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of training paths with status and approval information
Basic Scenario	1. Administrator navigates to the Training Paths section 2. System retrieves all training paths 3. System displays paths with title, author, status, and approval state 4. Administrator can filter paths by status or approval state 5. Administrator can view detailed training path information
Error Scenario	1. If no training paths exist, the system displays an empty-state message 2. If the training path list cannot be retrieved, the system shows an error notification

• Specification of the Use Case Approve Training Path

Table 1.88: Use Case Specification: Approve Training Path

Title	Approve Training Path
Summary	Administrator reviews a submitted training path and approves it for publication and possible featuring
Actor	Administrator
Precondition	Administrator is authenticated, verified, and authorized through the admin-only policy; a training path is pending review
Post-condition	The training path becomes approved and may later be featured or ordered among highlighted offerings
Basic Scenario	1. Administrator opens the training path approval dashboard 2. System lists submitted training paths 3. Administrator reviews pedagogical and operational information 4. Administrator confirms the approval action 5. System updates the training path status 6. The training path becomes available within approved public listings
Error Scenario	1. If the training path no longer satisfies review conditions, approval is refused 2. If the administrator lacks permission, the request is denied 3. If the update fails, the system retains the previous state and records the failure

- **Specification of the Use Case Reject Training Path (with feedback)**

Table 1.89: Use Case Specification: Reject Training Path (with feedback)

Title	Reject Training Path (with feedback)
Summary	Administrator rejects a submitted training path and provides feedback to the author
Actor	Administrator
Precondition	The administrator is authenticated and has selected a training path pending review
Post-condition	The training path is rejected and the author receives feedback for revisions
Basic Scenario	1. Administrator selects a training path pending review 2. Administrator clicks Reject 3. System displays the rejection form with feedback field 4. Administrator enters rejection feedback explaining the reasons 5. System updates the training path status to rejected 6. System sends a notification to the author with the feedback
Error Scenario	1. If the training path is already approved or rejected, the system displays a notice 2. If the rejection fails, the system displays an error and retains the current state

- **Specification of the Use Case Feature / Unfeature Path**

Table 1.90: Use Case Specification: Feature / Unfeature Path

Title	Feature / Unfeature Path
Summary	Administrator marks a training path as featured or removes the featured status
Actor	Administrator
Precondition	The administrator is authenticated and has selected an approved training path
Post-condition	The training path featured status is updated and it appears or disappears from featured listings
Basic Scenario	1. Administrator selects an approved training path 2. Administrator clicks Feature or Unfeature 3. System confirms the action 4. System updates the training path featured status 5. System confirms the update and displays the new status
Error Scenario	1. If the training path is not approved, the system prevents featuring 2. If the update fails, the system displays an error and retains the current state

- **Specification of the Use Case Reorder Featured Paths**

Table 1.91: Use Case Specification: Reorder Featured Paths

Title	Reorder Featured Paths
Summary	Administrator changes the display order of featured training paths on the landing page
Actor	Administrator
Precondition	The administrator is authenticated and has at least two featured training paths
Post-condition	The featured training paths are reordered and the landing page displays them in the new order
Basic Scenario	1. Administrator navigates to the Featured Paths section 2. System displays all featured training paths with their current order 3. Administrator drags and drops paths to reorder them 4. System updates the order values for all featured paths 5. System confirms the reorder and displays the new order
Error Scenario	1. If the reorder fails, the system displays an error and retains the previous order 2. If there are fewer than two featured paths, the system displays a notice

- **Specification of the Use Case List VM Assignment Queue**

Table 1.92: Use Case Specification: List VM Assignment Queue

Title	List VM Assignment Queue
Summary	Administrator views all pending VM assignment requests submitted by teachers
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of pending VM assignment requests with details and status
Basic Scenario	1. Administrator navigates to the VM Assignment Queue section 2. System retrieves all pending VM assignment requests 3. System displays requests with teacher, training path, unit, and requested template 4. Administrator can filter requests by teacher or training path 5. Administrator can view detailed request information
Error Scenario	1. If no pending requests exist, the system displays an empty-state message 2. If the request list cannot be retrieved, the system shows an error notification

- **Specification of the Use Case View Pending VM Assignments**

Table 1.93: Use Case Specification: View Pending VM Assignments

Title	View Pending VM Assignments
Summary	Administrator views detailed information about pending VM assignment requests
Actor	Administrator
Precondition	The administrator is authenticated and has selected a pending VM assignment request
Post-condition	The administrator views comprehensive details about the VM assignment request
Basic Scenario	1. Administrator selects a pending VM assignment request 2. System displays detailed request information 3. System shows teacher name, training path, unit, and requested template 4. System displays resource requirements and current infrastructure capacity 5. Administrator can approve or reject the request from this view
Error Scenario	1. If the request no longer exists, the system displays a not-found error 2. If the request details cannot be retrieved, the system shows an error notification

- **Specification of the Use Case Approve VM Assignment**

Table 1.94: Use Case Specification: Approve VM Assignment

Title	Approve VM Assignment
Summary	Administrator approves a VM assignment request, linking the template to the training unit
Actor	Administrator
Precondition	The administrator is authenticated and has selected a pending VM assignment request
Post-condition	The VM assignment is approved and the teacher is notified of the approval
Basic Scenario	1. Administrator selects a pending VM assignment request 2. Administrator reviews the request details and resource requirements 3. Administrator clicks Approve 4. System validates infrastructure capacity 5. System links the VM template to the training unit 6. System updates the request status to approved 7. System sends a notification to the teacher
Error Scenario	1. If the requested template no longer exists, the system flags the request as invalid 2. If infrastructure capacity is insufficient, the system suggests an alternative template

- **Specification of the Use Case Reject VM Assignment**

Table 1.95: Use Case Specification: Reject VM Assignment

Title	Reject VM Assignment
Summary	Administrator rejects a VM assignment request and provides feedback to the teacher
Actor	Administrator
Precondition	The administrator is authenticated and has selected a pending VM assignment request
Post-condition	The VM assignment is rejected and the teacher is notified with the rejection reason
Basic Scenario	1. Administrator selects a pending VM assignment request 2. Administrator clicks Reject 3. System displays the rejection form with feedback field 4. Administrator enters rejection feedback explaining the reasons 5. System updates the request status to rejected 6. System sends a notification to the teacher with the feedback
Error Scenario	1. If the request is already approved or rejected, the system displays a notice 2. If the rejection fails, the system displays an error and retains the current state

3.2.4.7 Refinement of the Use Case Approve VM Assignments

Figure 1.20 presents the refinement of the Use Case Approve VM Assignments for the Administrator actor. This use case includes the following sub-use cases:

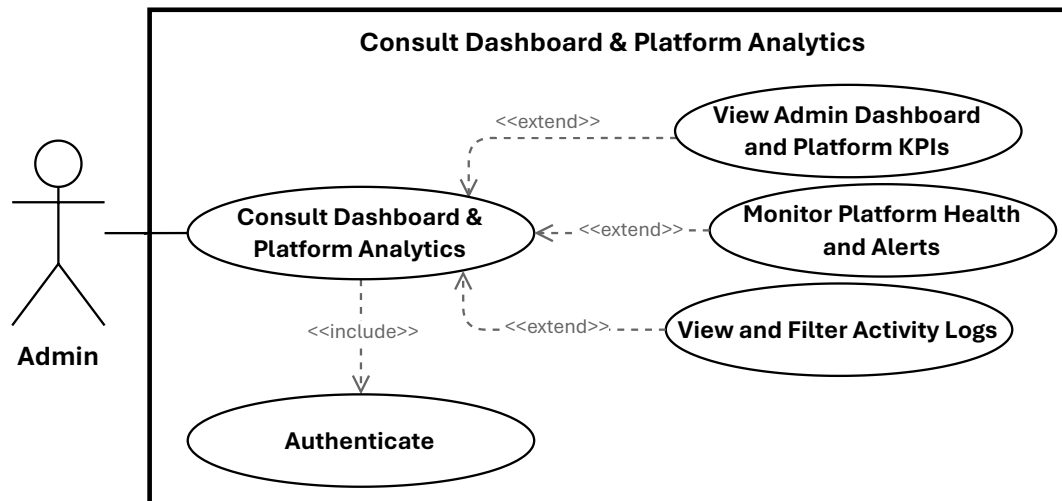


Figure 1.18: Use Case Approve VM Assignments

• Specification of the Use Case List All VM Assignments

Table 1.96: Use Case Specification: List All VM Assignments

Title	List All VM Assignments
Summary	Administrator views all VM template assignments to training units with their status
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of VM assignments with status and configuration details
Basic Scenario	1. Administrator navigates to the VM Assignments section 2. System retrieves all VM template assignments 3. System displays assignments with template, training path, unit, and status 4. Administrator can filter assignments by status or training path 5. Administrator can view detailed assignment information
Error Scenario	1. If no assignments exist, the system displays an empty-state message 2. If the assignment list cannot be retrieved, the system shows an error notification

• Specification of the Use Case View Pending VM Assignments (Duplicate)

Table 1.97: Use Case Specification: View Pending VM Assignments

Title	View Pending VM Assignments
Summary	Administrator views all pending VM assignment requests awaiting approval
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of pending VM assignments with teacher and template information
Basic Scenario	1. Administrator navigates to the Pending VM Assignments section 2. System retrieves all pending VM assignment requests 3. System displays assignments with teacher, training path, unit, and requested template 4. Administrator can filter assignments by teacher or training path 5. Administrator can approve or reject assignments from this view
Error Scenario	1. If no pending assignments exist, the system displays an empty-state message 2. If the assignment list cannot be retrieved, the system shows an error notification

- **Specification of the Use Case Approve VM Assignments**

Table 1.98: Use Case Specification: Approve VM Assignments

Title	Approve VM Assignments
Summary	Administrator approves pending VM assignment requests, linking templates to training units
Actor	Administrator
Precondition	The administrator is authenticated and has selected pending VM assignment requests
Post-condition	The VM assignments are approved and the teachers are notified of the approvals
Basic Scenario	1. Administrator selects one or more pending VM assignment requests 2. Administrator reviews the request details and resource requirements 3. Administrator clicks Approve 4. System validates infrastructure capacity for each request 5. System links the VM templates to the training units 6. System updates the request statuses to approved 7. System sends notifications to the teachers
Error Scenario	1. If any requested template no longer exists, the system flags that request as invalid 2. If infrastructure capacity is insufficient, the system suggests alternative templates

- **Specification of the Use Case Reject VM Assignments**

Table 1.99: Use Case Specification: Reject VM Assignments

Title	Reject VM Assignments
Summary	Administrator rejects pending VM assignment requests and provides feedback to teachers
Actor	Administrator
Precondition	The administrator is authenticated and has selected pending VM assignment requests
Post-condition	The VM assignments are rejected and the teachers are notified with rejection reasons
Basic Scenario	1. Administrator selects one or more pending VM assignment requests 2. Administrator clicks Reject 3. System displays the rejection form with feedback field 4. Administrator enters rejection feedback explaining the reasons 5. System updates the request statuses to rejected 6. System sends notifications to the teachers with the feedback
Error Scenario	1. If any request is already approved or rejected, the system displays a notice 2. If the rejection fails, the system displays an error and retains the current state

3.2.4.8 Refinement of the Use Case Manage Reservations

Figure 1.20 presents the refinement of the Use Case Manage Reservations for the Administrator actor. This use case includes the following sub-use cases:

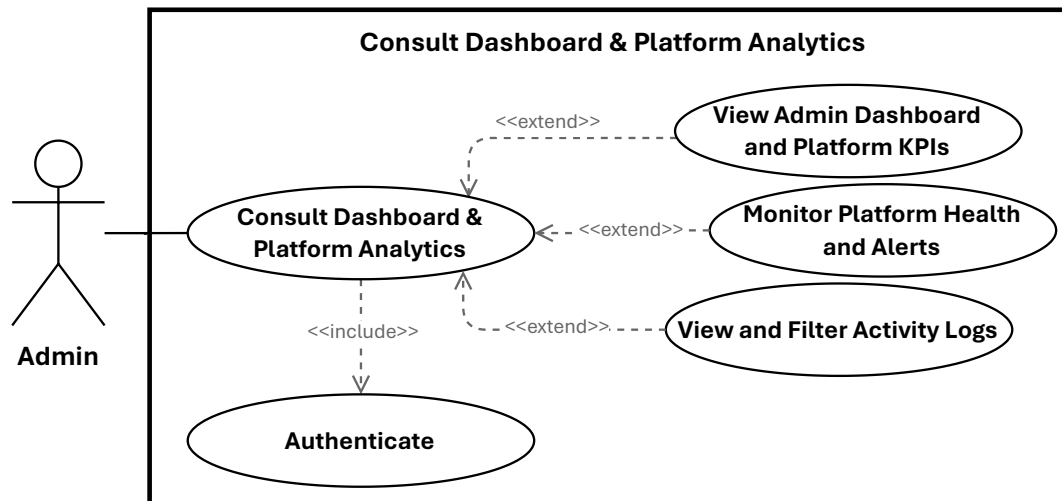


Figure 1.19: Use Case Manage Reservations

• Specification of the Use Case List All USB Reservations

Table 1.100: Use Case Specification: List All USB Reservations

Title	List All USB Reservations
Summary	Administrator views all USB device reservations with their status, user, and time slot
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of USB reservations with status and scheduling information
Basic Scenario	1. Administrator navigates to the USB Reservations section 2. System retrieves all USB device reservations 3. System displays reservations with device, user, status, and time slot 4. Administrator can filter reservations by status or device 5. Administrator can view detailed reservation information
Error Scenario	1. If no reservations exist, the system displays an empty-state message 2. If the reservation list cannot be retrieved, the system shows an error notification

• Specification of the Use Case View Pending Reservations

Table 1.101: Use Case Specification: View Pending Reservations

Title	View Pending Reservations
Summary	Administrator views all pending USB device reservation requests awaiting approval
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of pending reservations with user and device information
Basic Scenario	1. Administrator navigates to the Pending Reservations section 2. System retrieves all pending USB reservation requests 3. System displays reservations with device, user, and requested time slot 4. Administrator can filter reservations by device or user 5. Administrator can approve or reject reservations from this view
Error Scenario	1. If no pending reservations exist, the system displays an empty-state message 2. If the reservation list cannot be retrieved, the system shows an error notification

- **Specification of the Use Case View Upcoming Reservations**

Table 1.102: Use Case Specification: View Upcoming Reservations

Title	View Upcoming Reservations
Summary	Administrator views all approved USB device reservations scheduled for the future
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of upcoming reservations with scheduling details
Basic Scenario	1. Administrator navigates to the Upcoming Reservations section 2. System retrieves all approved future USB reservations 3. System displays reservations with device, user, and scheduled time slot 4. Administrator can filter reservations by date range or device 5. Administrator can view detailed reservation information
Error Scenario	1. If no upcoming reservations exist, the system displays an empty-state message 2. If the reservation list cannot be retrieved, the system shows an error notification

- **Specification of the Use Case Approve USB Reservation**

Table 1.103: Use Case Specification: Approve USB Reservation

Title	Approve USB Reservation
Summary	Administrator approves a pending USB device reservation request
Actor	Administrator
Precondition	The administrator is authenticated and has selected a pending USB reservation request
Post-condition	The USB reservation is approved and the engineer is notified of the approval
Basic Scenario	1. Administrator selects a pending USB reservation request 2. Administrator reviews the request details and device availability 3. Administrator clicks Approve 4. System validates device availability for the requested time slot 5. System updates the reservation status to approved 6. System sends a notification to the engineer
Error Scenario	1. If the device is already reserved for the time slot, the system suggests an alternative time 2. If the approval fails, the system displays an error and retains the pending state

- **Specification of the Use Case Reject USB Reservation**

Table 1.104: Use Case Specification: Reject USB Reservation

Title	Reject USB Reservation
Summary	Administrator rejects a pending USB device reservation request and provides feedback
Actor	Administrator
Precondition	The administrator is authenticated and has selected a pending USB reservation request
Post-condition	The USB reservation is rejected and the engineer is notified with the rejection reason
Basic Scenario	1. Administrator selects a pending USB reservation request 2. Administrator clicks Reject 3. System displays the rejection form with feedback field 4. Administrator enters rejection feedback explaining the reasons 5. System updates the reservation status to rejected 6. System sends a notification to the engineer with the feedback
Error Scenario	1. If the reservation is already approved or rejected, the system displays a notice 2. If the rejection fails, the system displays an error and retains the current state

- **Specification of the Use Case Block USB Device Time Slot**

Table 1.105: Use Case Specification: Block USB Device Time Slot

Title	Block USB Device Time Slot
Summary	Administrator blocks a specific time slot for a USB device to prevent reservations during maintenance
Actor	Administrator
Precondition	The administrator is authenticated and has selected a USB device
Post-condition	The USB device time slot is blocked and no reservations can be made for that period
Basic Scenario	1. Administrator selects a USB device from the device list 2. Administrator clicks Block Time Slot 3. System displays the time slot blocking form 4. Administrator selects the date range and time to block 5. Administrator enters a reason for the block 6. System creates a blocked time slot for the device 7. System confirms the block and displays the blocked periods
Error Scenario	1. If the time slot conflicts with existing reservations, the system displays a conflict warning 2. If the block creation fails, the system displays an error and retains the current state

- **Specification of the Use Case List Camera Reservations**

Table 1.106: Use Case Specification: List Camera Reservations

Title	List Camera Reservations
Summary	Administrator views all camera reservations with their status, user, and time slot
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of camera reservations with status and scheduling information
Basic Scenario	1. Administrator navigates to the Camera Reservations section 2. System retrieves all camera reservations 3. System displays reservations with camera, user, status, and time slot 4. Administrator can filter reservations by status or camera 5. Administrator can view detailed reservation information
Error Scenario	1. If no reservations exist, the system displays an empty-state message 2. If the reservation list cannot be retrieved, the system shows an error notification

- **Specification of the Use Case View Pending Camera Reservations**

Table 1.107: Use Case Specification: View Pending Camera Reservations

Title	View Pending Camera Reservations
Summary	Administrator views all pending camera reservation requests awaiting approval
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of pending camera reservations with user and camera information
Basic Scenario	1. Administrator navigates to the Pending Camera Reservations section 2. System retrieves all pending camera reservation requests 3. System displays reservations with camera, user, and requested time slot 4. Administrator can filter reservations by camera or user 5. Administrator can approve or reject reservations from this view
Error Scenario	1. If no pending reservations exist, the system displays an empty-state message 2. If the reservation list cannot be retrieved, the system shows an error notification

- **Specification of the Use Case Approve Camera Reservation**

Table 1.108: Use Case Specification: Approve Camera Reservation

Title	Approve Camera Reservation
Summary	Administrator approves a pending camera reservation request
Actor	Administrator
Precondition	The administrator is authenticated and has selected a pending camera reservation request
Post-condition	The camera reservation is approved and the engineer is notified of the approval
Basic Scenario	1. Administrator selects a pending camera reservation request 2. Administrator reviews the request details and camera availability 3. Administrator clicks Approve 4. System validates camera availability for the requested time slot 5. System updates the reservation status to approved 6. System sends a notification to the engineer
Error Scenario	1. If the camera is already reserved for the time slot, the system suggests an alternative time 2. If the approval fails, the system displays an error and retains the pending state

- **Specification of the Use Case Reject Camera Reservation**

Table 1.109: Use Case Specification: Reject Camera Reservation

Title	Reject Camera Reservation
Summary	Administrator rejects a pending camera reservation request and provides feedback
Actor	Administrator
Precondition	The administrator is authenticated and has selected a pending camera reservation request
Post-condition	The camera reservation is rejected and the engineer is notified with the rejection reason
Basic Scenario	1. Administrator selects a pending camera reservation request 2. Administrator clicks Reject 3. System displays the rejection form with feedback field 4. Administrator enters rejection feedback explaining the reasons 5. System updates the reservation status to rejected 6. System sends a notification to the engineer with the feedback
Error Scenario	1. If the reservation is already approved or rejected, the system displays a notice 2. If the rejection fails, the system displays an error and retains the current state

- **Specification of the Use Case Block Camera Time Slot**

Table 1.110: Use Case Specification: Block Camera Time Slot

Title	Block Camera Time Slot
Summary	Administrator blocks a specific time slot for a camera to prevent reservations during maintenance
Actor	Administrator
Precondition	The administrator is authenticated and has selected a camera
Post-condition	The camera time slot is blocked and no reservations can be made for that period
Basic Scenario	1. Administrator selects a camera from the camera list 2. Administrator clicks Block Time Slot 3. System displays the time slot blocking form 4. Administrator selects the date range and time to block 5. Administrator enters a reason for the block 6. System creates a blocked time slot for the camera 7. System confirms the block and displays the blocked periods
Error Scenario	1. If the time slot conflicts with existing reservations, the system displays a conflict warning 2. If the block creation fails, the system displays an error and retains the current state

3.2.4.9 Refinement of the Use Case Manage Payout & Refunds

Figure 1.20 presents the refinement of the Use Case Manage Payout & Refunds for the Administrator actor. This use case includes the following sub-use cases:

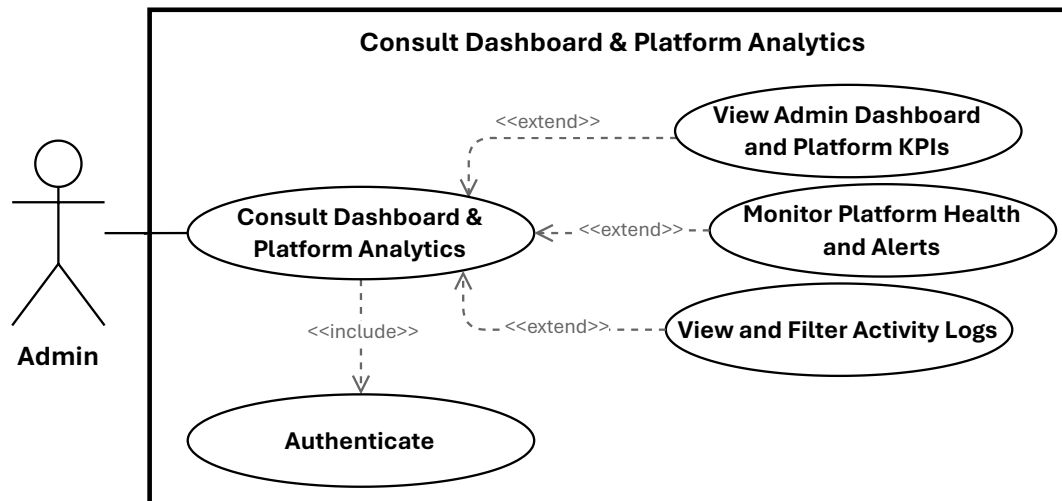


Figure 1.20: Use Case Manage Payout & Refunds

• Specification of the Use Case List Payout Requests

Table 1.111: Use Case Specification: List Payout Requests

Title	List Payout Requests
Summary	Administrator views all payout requests submitted by teachers with their status and amounts
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of payout requests with status, teacher, and amount information
Basic Scenario	1. Administrator navigates to the Payout Requests section 2. System retrieves all payout requests 3. System displays requests with teacher, amount, status, and request date 4. Administrator can filter requests by status or teacher 5. Administrator can view detailed payout request information
Error Scenario	1. If no payout requests exist, the system displays an empty-state message 2. If the payout list cannot be retrieved, the system shows an error notification

• Specification of the Use Case Export Payouts Report

Table 1.112: Use Case Specification: Export Payouts Report

Title	Export Payouts Report
Summary	Administrator exports a report of payout requests for accounting or analysis purposes
Actor	Administrator
Precondition	The administrator is authenticated and has selected payout requests to export
Post-condition	A payout report file is generated and downloaded in the requested format
Basic Scenario	1. Administrator navigates to the Payout Requests section 2. Administrator applies filters for the report (date range, status, teacher) 3. Administrator clicks Export Report 4. System generates the report with selected payout data 5. System offers download options: CSV, PDF, or Excel 6. Administrator selects the format and downloads the report
Error Scenario	1. If no data matches the filters, the system displays an empty-state message 2. If the report generation fails, the system shows an error notification

- **Specification of the Use Case Approve Payout Request**

Table 1.113: Use Case Specification: Approve Payout Request

Title	Approve Payout Request
Summary	Administrator approves a pending payout request for a teacher
Actor	Administrator
Precondition	The administrator is authenticated and has selected a pending payout request
Post-condition	The payout request is approved and the teacher is notified of the approval
Basic Scenario	1. Administrator selects a pending payout request 2. Administrator reviews the payout details and teacher earnings history 3. Administrator clicks Approve 4. System validates the payout amount and teacher account 5. System updates the payout status to approved 6. System sends a notification to the teacher
Error Scenario	1. If the payout request is already approved or rejected, the system displays a notice 2. If the approval fails, the system displays an error and retains the current state

- **Specification of the Use Case Reject Payout Request**

Table 1.114: Use Case Specification: Reject Payout Request

Title	Reject Payout Request
Summary	Administrator rejects a pending payout request and provides feedback to the teacher
Actor	Administrator
Precondition	The administrator is authenticated and has selected a pending payout request
Post-condition	The payout request is rejected and the teacher is notified with the rejection reason
Basic Scenario	1. Administrator selects a pending payout request 2. Administrator clicks Reject 3. System displays the rejection form with feedback field 4. Administrator enters rejection feedback explaining the reasons 5. System updates the payout status to rejected 6. System sends a notification to the teacher with the feedback
Error Scenario	1. If the payout request is already approved or rejected, the system displays a notice 2. If the rejection fails, the system displays an error and retains the current state

- **Specification of the Use Case Process / Transfer Payout**

Table 1.115: Use Case Specification: Process / Transfer Payout

Title	Process / Transfer Payout
Summary	Administrator initiates the actual transfer of funds for an approved payout request
Actor	Administrator
Precondition	The administrator is authenticated and has selected an approved payout request
Post-condition	The payout transfer is initiated and the payout status is updated to processing or completed
Basic Scenario	1. Administrator selects an approved payout request 2. Administrator clicks Process Payout 3. System displays the payout details and transfer options 4. Administrator confirms the transfer 5. System initiates the transfer through the payment processor 6. System updates the payout status to processing 7. System logs the transfer for audit purposes
Error Scenario	1. If the payout is not approved, the system prevents processing 2. If the transfer fails, the system displays an error and marks the payout for manual review

- **Specification of the Use Case List Refund Requests**

Table 1.116: Use Case Specification: List Refund Requests

Title	List Refund Requests
Summary	Administrator views all refund requests submitted by engineers with their status and amounts
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a list of refund requests with status, engineer, and amount information
Basic Scenario	1. Administrator navigates to the Refund Requests section 2. System retrieves all refund requests 3. System displays requests with engineer, amount, status, and request date 4. Administrator can filter requests by status or engineer 5. Administrator can view detailed refund request information
Error Scenario	1. If no refund requests exist, the system displays an empty-state message 2. If the refund list cannot be retrieved, the system shows an error notification

- **Specification of the Use Case View All Refunds History**

Table 1.117: Use Case Specification: View All Refunds History

Title	View All Refunds History
Summary	Administrator views the complete history of all refund transactions processed on the platform
Actor	Administrator
Precondition	The administrator is authenticated with admin privileges
Post-condition	The administrator views a comprehensive list of all refunds with status and transaction details
Basic Scenario	1. Administrator navigates to the Refunds History section 2. System retrieves all refund transactions 3. System displays refunds with engineer, amount, status, and processing date 4. Administrator can filter refunds by status, date range, or engineer 5. Administrator can view detailed refund transaction information
Error Scenario	1. If no refunds exist, the system displays an empty-state message 2. If the refund history cannot be retrieved, the system shows an error notification

- **Specification of the Use Case Approve Refund Request**

Table 1.118: Use Case Specification: Approve Refund Request

Title	Approve Refund Request
Summary	Administrator approves a pending refund request and initiates the Stripe refund
Actor	Administrator
Precondition	The administrator is authenticated and has selected a pending refund request
Post-condition	The refund request is approved and the Stripe refund is initiated
Basic Scenario	1. Administrator selects a pending refund request 2. Administrator reviews the refund details and payment information 3. Administrator clicks Approve 4. System validates the refund eligibility and amount 5. System updates the refund status to approved 6. System initiates the Stripe refund through the API 7. System sends a notification to the engineer
Error Scenario	1. If the refund request is already approved or rejected, the system displays a notice 2. If the Stripe refund fails, the system logs the error and marks the refund for manual processing

- **Specification of the Use Case Reject Refund Request**

Table 1.119: Use Case Specification: Reject Refund Request

Title	Reject Refund Request
Summary	Administrator rejects a pending refund request and provides feedback to the engineer
Actor	Administrator
Precondition	The administrator is authenticated and has selected a pending refund request
Post-condition	The refund request is rejected and the engineer is notified with the rejection reason
Basic Scenario	1. Administrator selects a pending refund request 2. Administrator clicks Reject 3. System displays the rejection form with feedback field 4. Administrator enters rejection feedback explaining the reasons 5. System updates the refund status to rejected 6. System sends a notification to the engineer with the feedback
Error Scenario	1. If the refund request is already approved or rejected, the system displays a notice 2. If the rejection fails, the system displays an error and retains the current state

- **Specification of the Use Case Execute Stripe Refund**

Table 1.120: Use Case Specification: Execute Stripe Refund

Title	Execute Stripe Refund
Summary	Administrator manually executes a Stripe refund for an approved refund request
Actor	Administrator
Precondition	The administrator is authenticated and has selected an approved refund request requiring manual processing
Post-condition	The Stripe refund is executed and the refund status is updated to completed or failed
Basic Scenario	1. Administrator selects an approved refund request marked for manual processing 2. Administrator clicks Execute Stripe Refund 3. System displays the refund details and original payment information 4. Administrator confirms the refund execution 5. System calls the Stripe API to execute the refund 6. System updates the refund status based on the result 7. System logs the refund execution for audit purposes
Error Scenario	1. If the Stripe API call fails, the system displays an error and marks the refund for retry 2. If the original payment cannot be refunded, the system displays an error with details

4 Sequence Diagrams

Sequence diagrams explain the chronological interaction among actors, controllers, services, models, and infrastructure components. In the current platform, the most important sequences are not limited to standard web CRUD operations; they also include payment confirmation, virtual lab provisioning, and controlled interaction with physical resources.

4.1 Training Path Enrollment and Checkout Sequence

This sequence begins when an engineer decides to access a paid or restricted training path. The user selects the training path, initiates checkout, and is redirected to the external payment flow when necessary. After Stripe confirms payment through the webhook endpoint, the platform updates the payment state and makes the related training path accessible through the learner's

workspace. The process can be summarized as follows:

1. Engineer selects a training path and initiates checkout
2. Checkout controller creates the payment context
3. External payment processor handles transaction validation
4. Stripe webhook returns event data to the platform
5. System validates the signature and records the webhook event
6. Payment status is updated to completed or failed
7. Enrollment access becomes available to the engineer
8. The engineer can later consult payment history and request refund where applicable

Figure 1.21 illustrates the complete enrollment and checkout sequence, including the interaction between the frontend, the Laravel backend, and the Stripe payment processor.

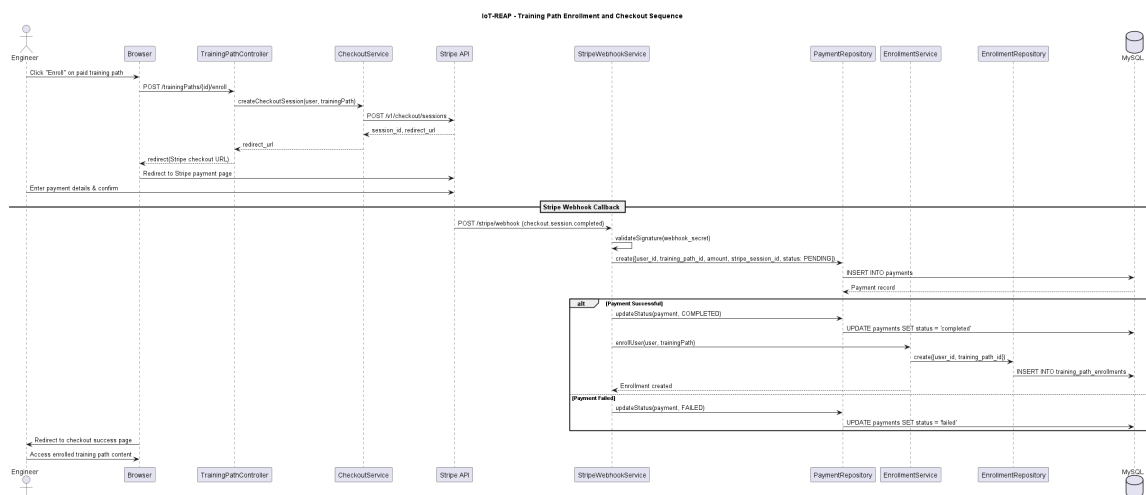


Figure 1.21: Sequence Diagram: Training Path Enrollment and Checkout

4.2 Teacher Submission for Review Sequence

This sequence models the transition from content authoring to publication governance. A teacher edits the training path and associated learning units, then submits the path for review. The platform validates ownership and readiness, changes the review status, and exposes the path to the administrator approval interface.

1. Teacher opens the teaching dashboard
2. Teacher updates the training path structure and content
3. Teacher triggers the submission action
4. Controller validates role, ownership, and submission conditions
5. Training path state changes to pending review
6. Administrator later retrieves the submission from the administrative dashboard

Figure 1.22 shows the teacher submission for review sequence, from content editing through the review status transition.

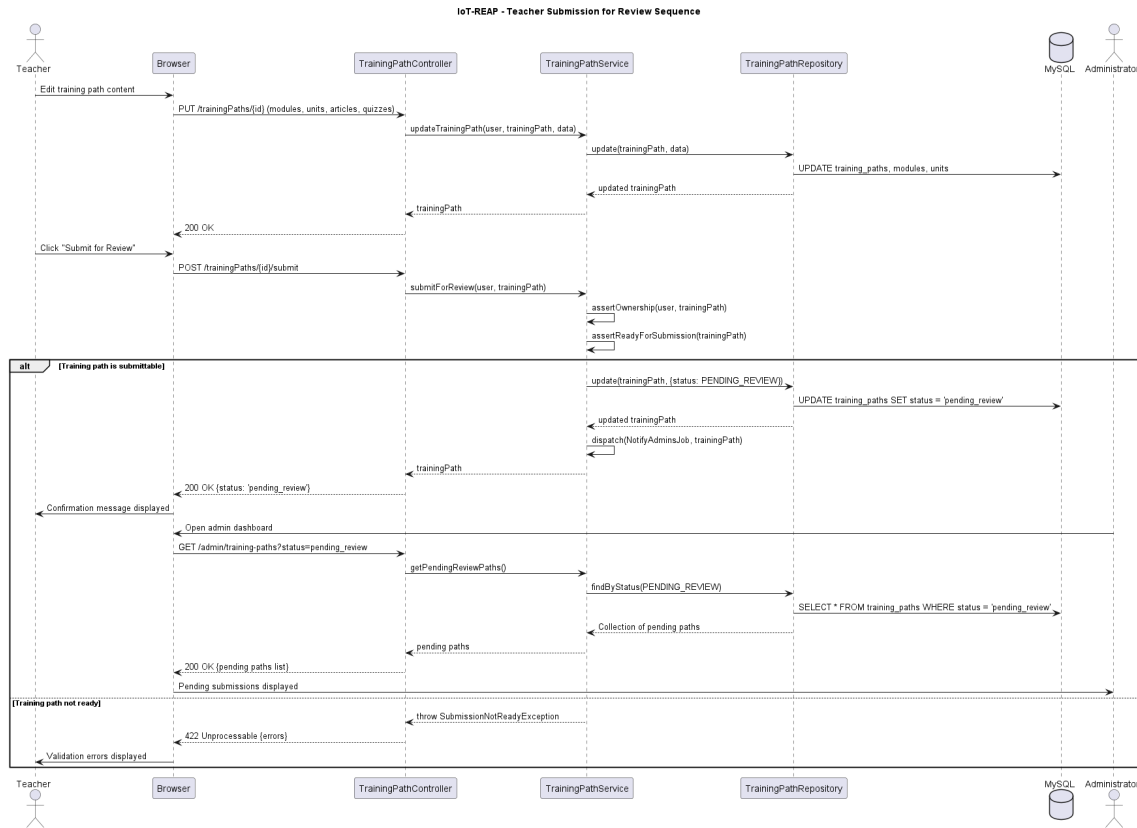


Figure 1.22: Sequence Diagram: Teacher Submission for Review

4.3 Virtual Machine Provisioning Sequence

The virtual machine provisioning sequence is a central differentiator of the platform. It links the user interface to infrastructure models and remote access capabilities.

1. Engineer sends a session creation request
2. Session controller checks authentication, verification, and provisioning permission
3. System evaluates available Proxmox servers and nodes
4. A VM session is created and associated with infrastructure metadata
5. The engineer consults the created session
6. A Guacamole token is generated on demand for remote access
7. The engineer may extend or terminate the session later

Figure 1.23 presents the virtual machine provisioning sequence, showing the interaction between the engineer, the session controller, the Proxmox API, and the Guacamole remote access gateway.

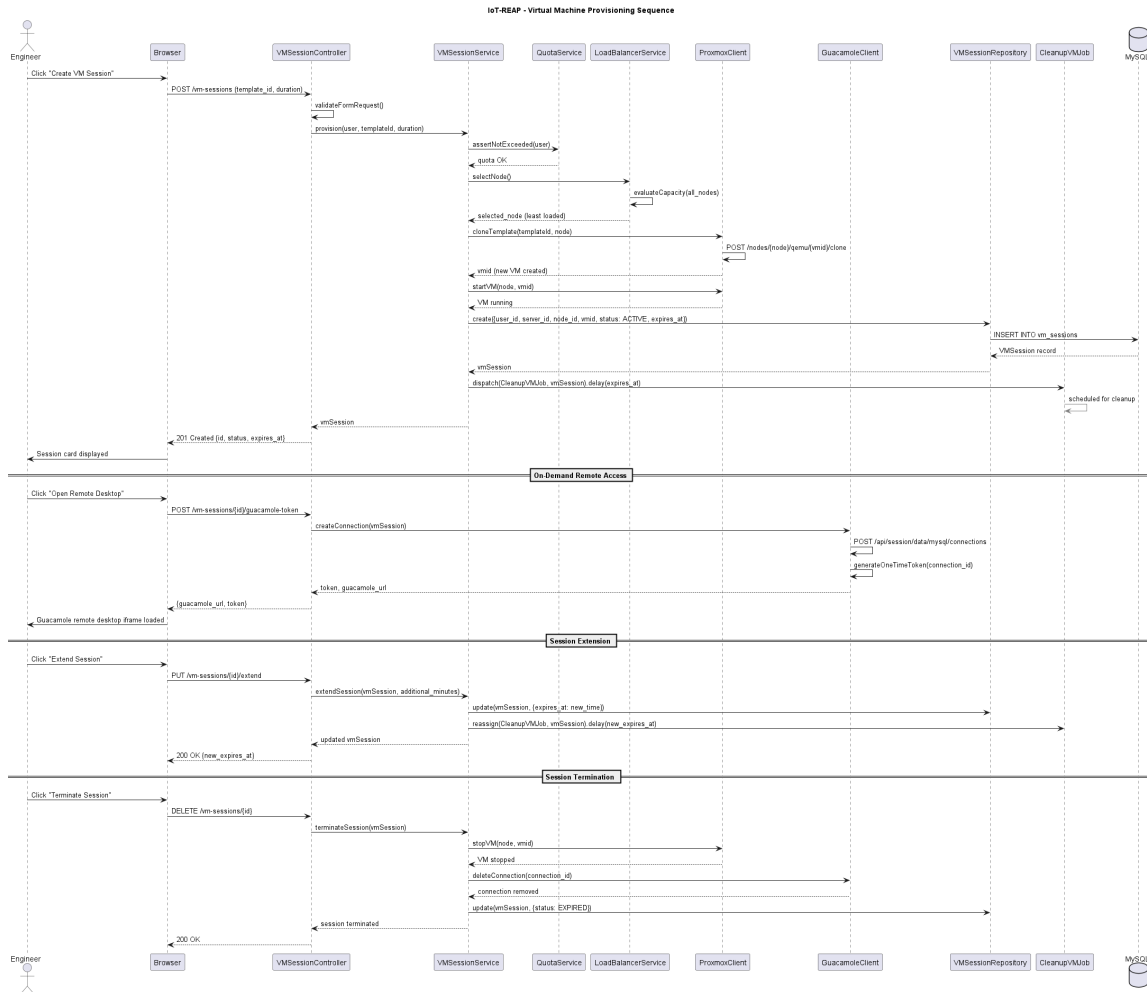


Figure 1.23: Sequence Diagram: Virtual Machine Provisioning

4.4 Hardware Attachment and Queue Sequence

This sequence governs access to USB devices attached to laboratory sessions. Because some devices are exclusive or dedicated, the workflow includes queueing behavior.

1. Engineer opens the hardware page associated with an active session
2. System returns visible devices and their current state
3. Engineer requests attachment of a selected device
4. System checks whether the device is available, attached, reserved, or dedicated
5. If the device is available, the attachment is executed
6. If the device is busy, the engineer may join a waiting queue
7. When the device becomes available, queue state is updated and notification can be issued

Figure 1.24 illustrates the hardware attachment and queue sequence, including the decision logic for device availability and the waiting queue mechanism.

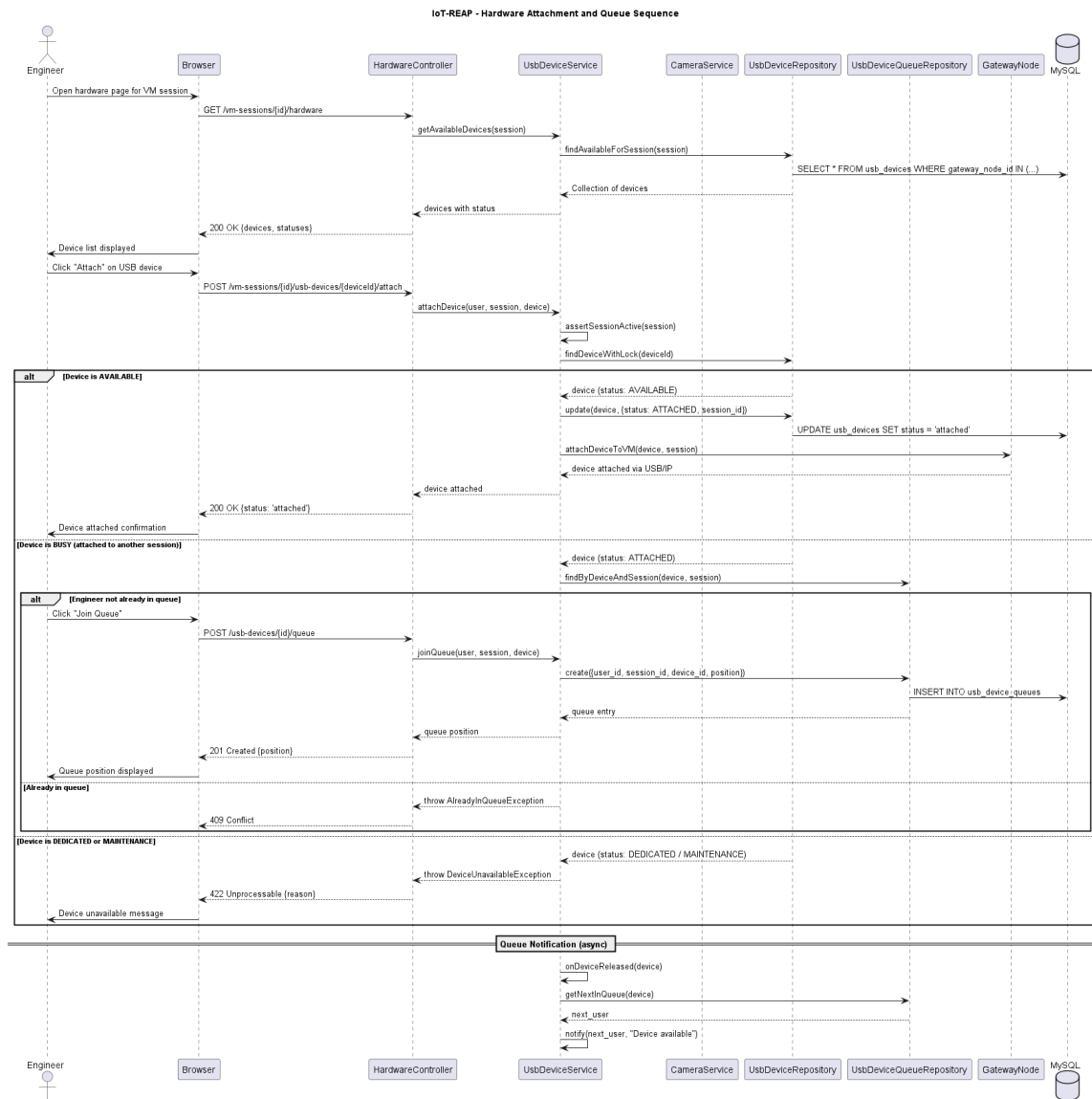


Figure 1.24: Sequence Diagram: Hardware Attachment and Queue

4.5 Quiz Attempt Submission Sequence

This sequence models the educational evaluation workflow.

1. Engineer opens the quiz associated with a training unit
2. System verifies attempt eligibility
3. A quiz attempt is created
4. Engineer submits answers
5. The platform validates and scores the attempt
6. Attempt history and completion status are updated
7. Training unit progress may be updated to reflect quiz completion or pass state

Figure 1.25 shows the quiz attempt submission sequence, from quiz opening through answer

submission, scoring, and progress update.

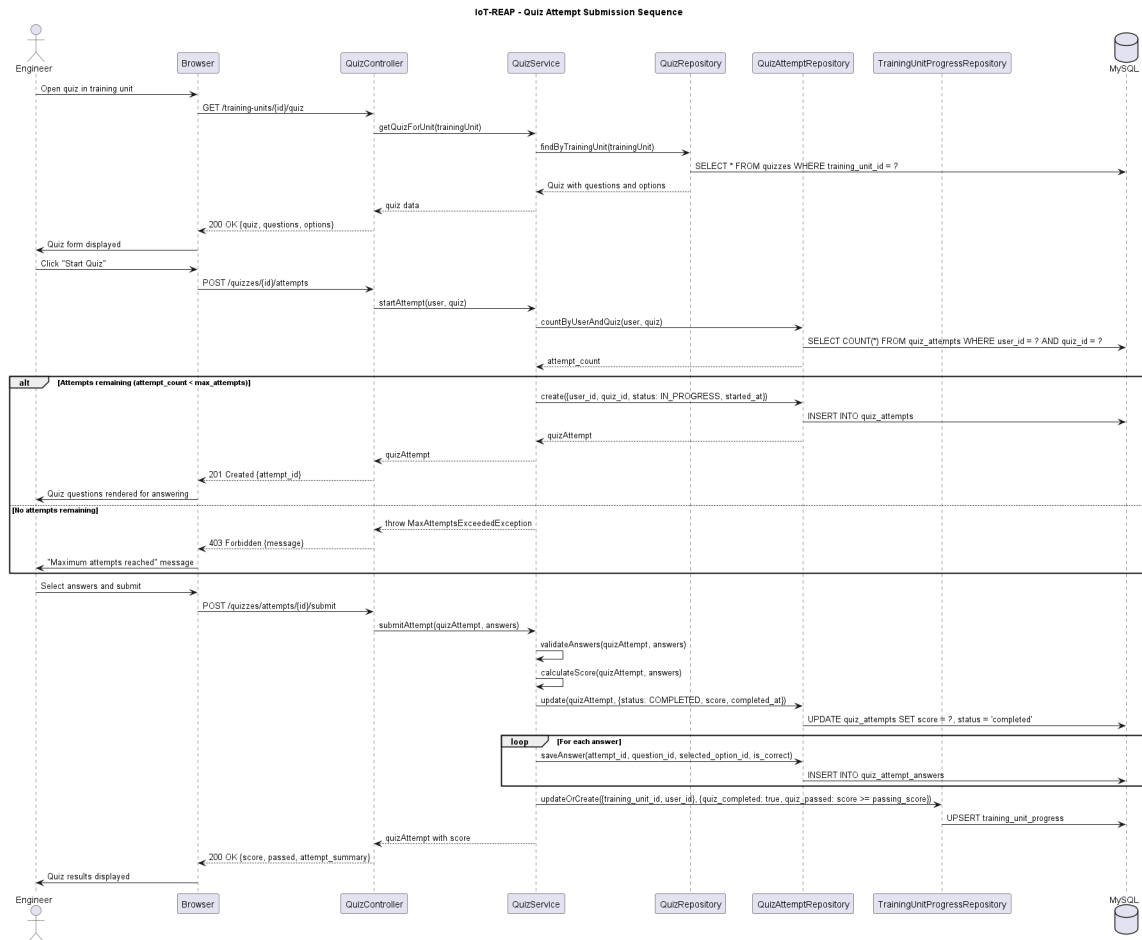


Figure 1.25: Sequence Diagram: Quiz Attempt Submission

4.6 Google OAuth Authentication Sequence

This sequence describes the social login flow when a user chooses to authenticate through Google OAuth. The process leverages Laravel Socialite to redirect the user to Google's consent screen, receive the authorization callback, and either link the Google account to an existing user or create a new account automatically. Role selection is presented when a new account is created through this pathway.

1. Guest clicks the Google login button on the authentication page
2. Laravel Socialite redirects the user to the Google OAuth consent screen
3. User authenticates with Google and grants permission
4. Google redirects back to the platform callback endpoint with an authorization code
5. Socialite exchanges the code for an access token and retrieves user profile data
6. System checks whether the Google account is already linked to an existing user
7. If linked, the user is authenticated and redirected to their dashboard

8. If not linked, a new account is created and the user is prompted to select a role
9. User selects a role and the account is finalized with the chosen role assignment

Figure 1.26 illustrates the Google OAuth authentication sequence, showing the interaction between the user, the Laravel application, and the Google identity provider.

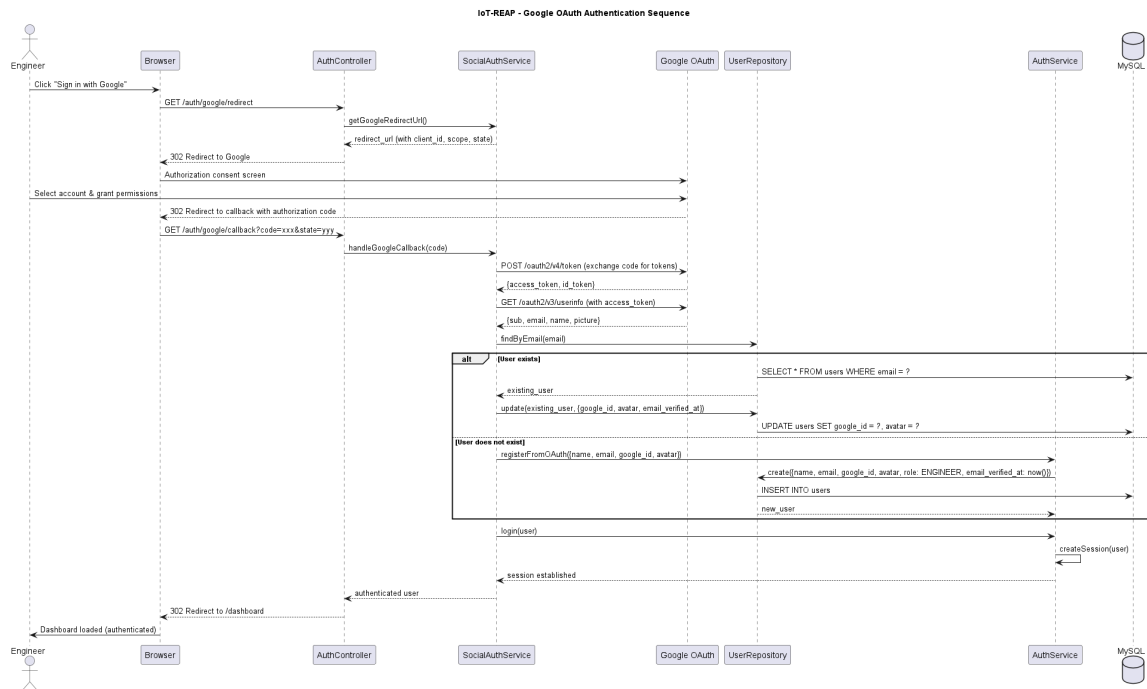


Figure 1.26: Sequence Diagram: Google OAuth Authentication

4.7 MQTT Robot Control Sequence

This sequence models the real-time command and telemetry exchange between an engineer and a robot connected through the MQTT broker. The engineer issues movement or action commands via the platform interface, which are published to the robot's command topic. The robot executes the command and publishes telemetry data back, which the platform subscribes to and relays to the engineer's dashboard.

1. Engineer opens the robot control interface within an active VM session
2. System verifies session ownership and robot access authorization
3. Engineer issues a command through the control interface
4. Platform publishes the command to the MQTT topic `robot/{robot_id}/commands` with QoS 1
5. Mosquitto broker delivers the command to the subscribed ESP32 controller
6. Robot executes the command and publishes telemetry to `robot/{robot_id}/telemetry`
7. Platform MQTT subscriber receives the telemetry update

8. Telemetry data is relayed to the engineer's interface through WebSocket or polling
9. Engineer observes the updated robot state and may issue further commands

Figure 1.27 presents the MQTT robot control sequence, showing the publish-subscribe interaction between the engineer, the Laravel backend, the Mosquitto broker, and the ESP32-controlled robot.

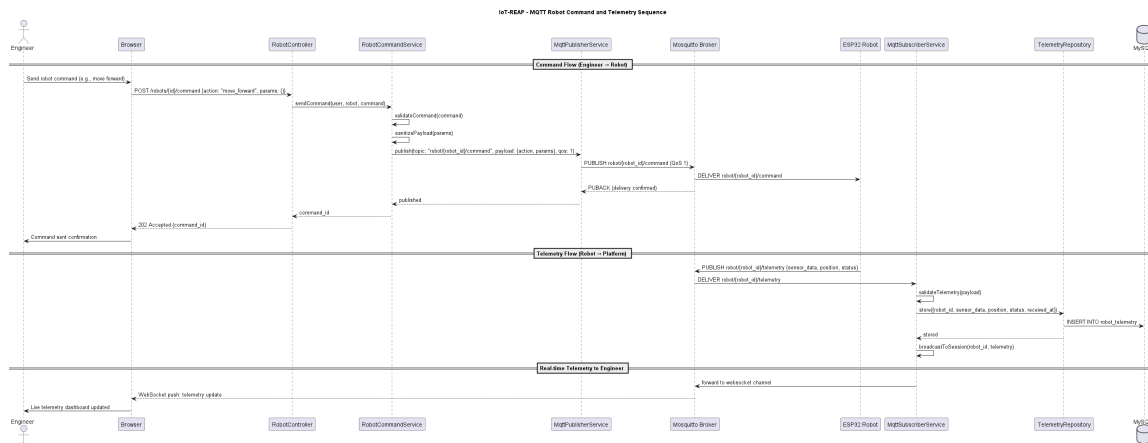


Figure 1.27: Sequence Diagram: MQTT Robot Control

4.8 Camera PTZ Control Sequence

This sequence describes how an engineer requests and exercises pan-tilt-zoom control over a PTZ-enabled camera within an active VM session. The workflow includes control request arbitration, exclusive session control creation, movement command dispatch, and control release when the engineer finishes or the session ends.

1. Engineer opens the camera stream view within an active session
2. System displays the live camera feed and available PTZ controls
3. Engineer requests camera control
4. System checks whether the camera is currently controlled by another user
5. If available, a CameraSessionControl record is created granting exclusive control
6. Engineer issues PTZ movement commands through the interface
7. System translates each command into camera-specific movement parameters
8. Camera executes the movement and the updated frame is reflected in the stream
9. Engineer may change stream resolution during the session
10. When the engineer releases control or the session ends, the control record is removed

Figure 1.28 illustrates the camera PTZ control sequence, from control acquisition through movement commands to control release.

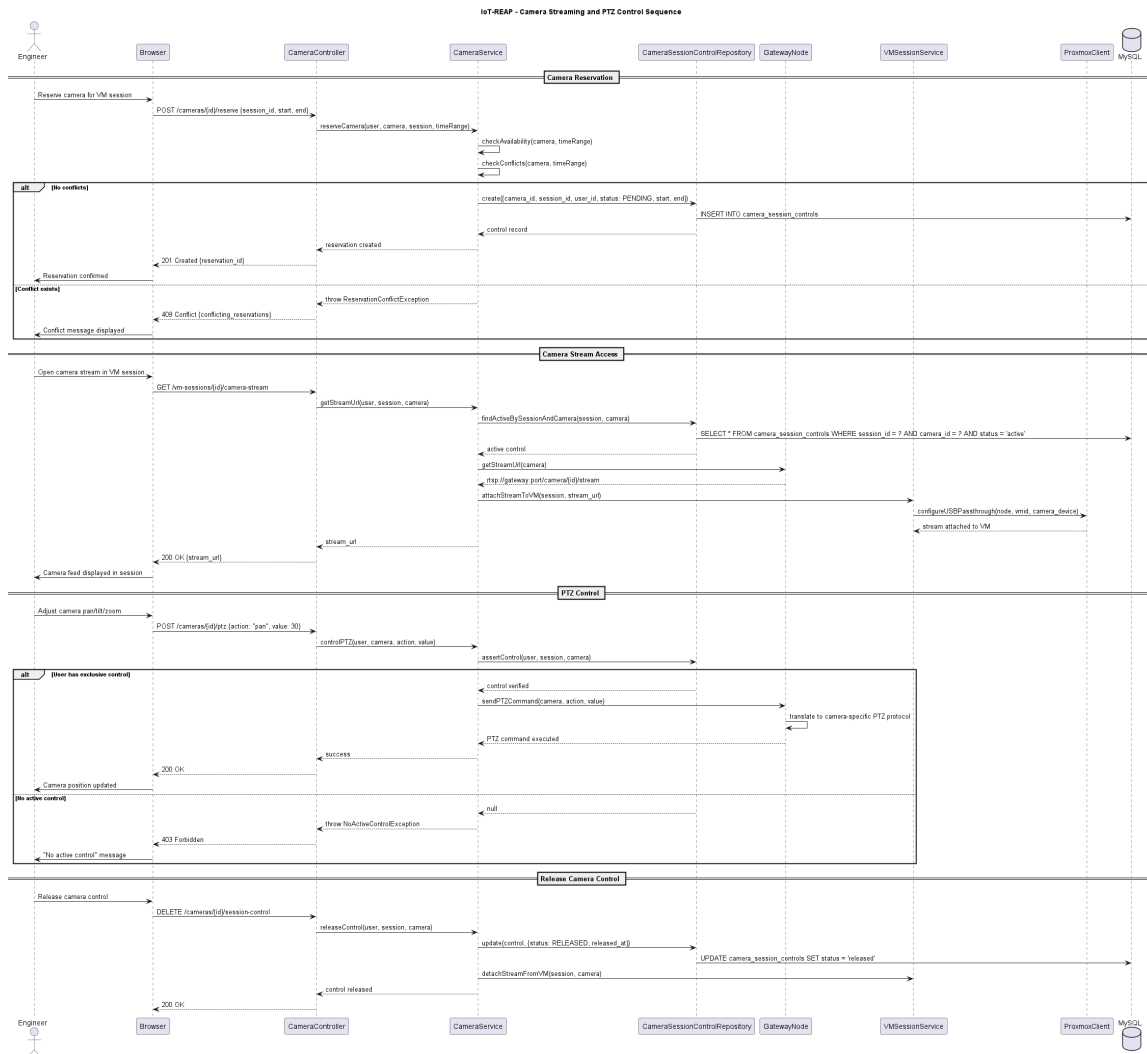


Figure 1.28: Sequence Diagram: Camera PTZ Control

4.9 Refund Processing Sequence

This sequence models the refund workflow from the engineer's initial request through administrative review and potential Stripe refund execution. It demonstrates how payment governance is handled as a multi-step auditable process rather than a simple transaction reversal.

1. Engineer submits a refund request for a completed payment
2. System validates that the payment exists and belongs to the requesting user
3. A RefundRequest record is created with pending status
4. Administrator opens the refund review interface
5. Administrator reviews the payment context and refund reason
6. Administrator approves or rejects the refund request
7. If approved, the system initiates a refund through the Stripe API
8. Stripe processes the refund and returns the result

9. System updates the refund request status and the payment state accordingly
10. Engineer is notified of the refund decision and outcome

Figure 1.29 presents the refund processing sequence, showing the interaction between the engineer, the administrator, the Laravel backend, and the Stripe refund API.

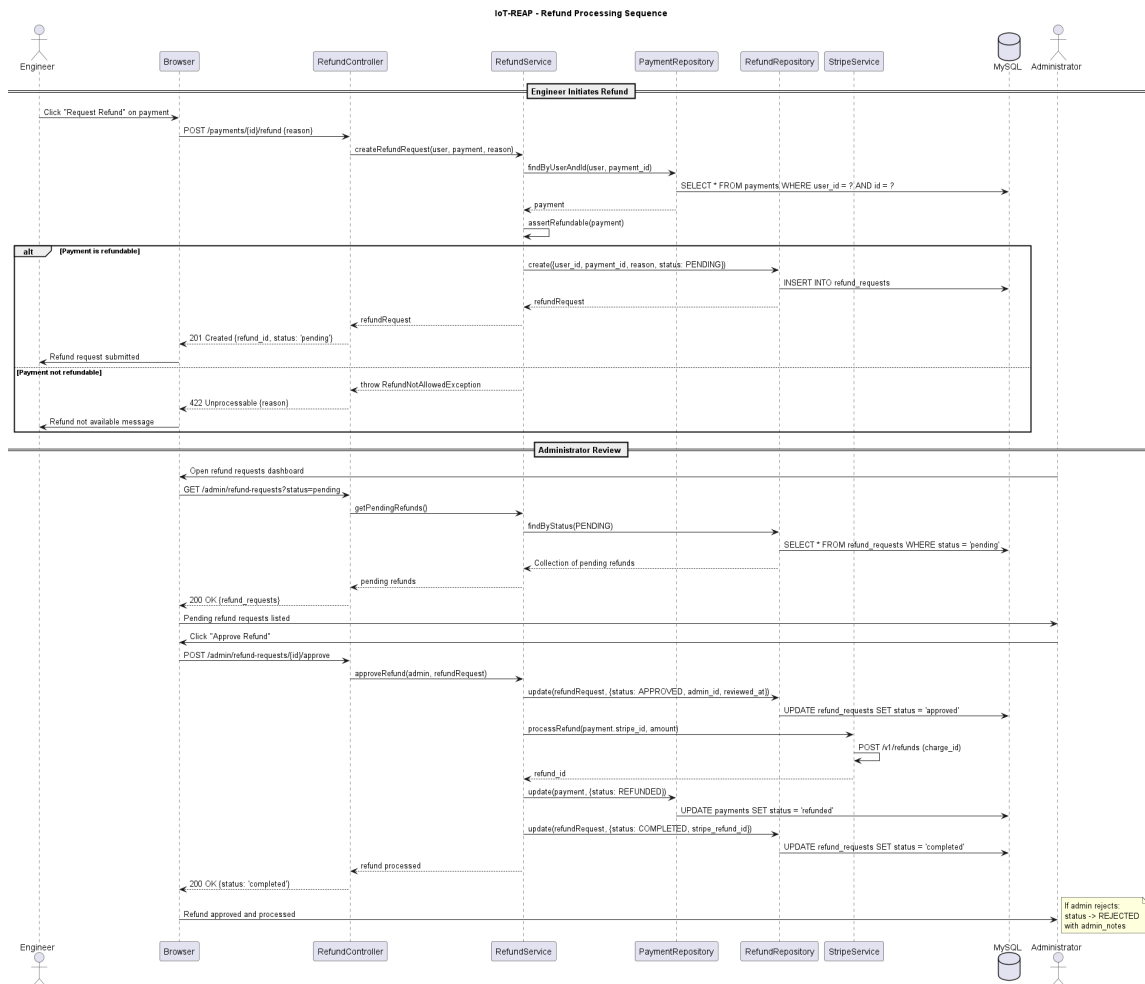


Figure 1.29: Sequence Diagram: Refund Processing

5 Class Diagram

The class diagram of the current platform is broader than that of a conventional learning portal because it combines academic content, learner engagement, payment processing, virtual machine infrastructure, device reservation, and operational monitoring. The conceptual model can therefore be organized into interrelated domains.

5.1 User and Access Domain

User Class: The `User` class is the central identity entity of the platform. It stores authentication and profile information, enforces role-aware behavior through helper methods such as `teacher` and `administrator` checks, and links a person to enrollments, taught training paths, quiz attempts, payments, and notifications. It is the pivot between educational participation and infrastructure usage.

Notification and Activity Classes: The `Notification`, `ActivityLog`, and `SystemAlert` classes support communication, auditability, and operational supervision. Notifications deliver user-level updates, activity logs record significant actions, and system alerts capture infrastructure or platform incidents that require administrative attention.

5.2 Learning Content Domain

TrainingPath Class: The `TrainingPath` class is the main pedagogical aggregate. It belongs to an instructor, contains modules and training units, receives enrollments, collects reviews and payments, and exposes status-dependent scopes such as `approved` or `pending review`. It represents the complete educational offering published on the platform.

TrainingPathModule and TrainingUnit Classes: `TrainingPathModule` structures a training path into ordered thematic groups, while `TrainingUnit` represents the granular learning unit. A training unit can contain one article, one video, one quiz, one virtual machine assignment, and multiple forms of progress tracking. These classes define the educational hierarchy used by both teachers and learners.

Article, Video, and Caption Classes: `Article` stores textual educational content and derived metrics such as word count and estimated reading time. `Video` stores media-related metadata, processing status, and streaming access paths. `Caption` complements the video model with subtitle and accessibility support. Together, these classes model the multimedia core of the learning experience.

Quiz Domain Classes: The assessment subsystem is formed by `Quiz`, `QuizQuestion`, `QuizQuestionOption`, `QuizAttempt`, and `QuizAttemptAnswer`. These classes represent the authored evaluation structure, learner answers, scoring context, attempt limits, and result history. They support both teacher-side authoring and learner-side evaluation.

Progress and Participation Classes: `TrainingUnitProgress`, `VideoProgress`, `TrainingUnitNote`, `TrainingPathEnrollment`, `TrainingPathReview`, `DiscussionThread`, `ThreadReply`, and `ThreadVote` model learner participation. They collectively express completion state, note taking, enrollment membership, reputation, peer interaction, and social learning dynamics.

Certificate Class: The `Certificate` class materializes completion outcomes. It belongs to a user and a training path, and provides verification and download URLs that support public authenticity checks.

5.3 Commerce and Revenue Domain

Payment and RefundRequest Classes: The `Payment` class records financial transactions associated with users and training paths. It stores amount-related data, completion state, and refund relationships. `RefundRequest` adds the post-payment governance layer by representing approval, rejection, processing, and failure states for refund handling.

PayoutRequest Class: The `PayoutRequest` class represents teacher-side revenue withdrawal requests. It links the requesting user with administrative review and processing decisions, thereby connecting platform earnings to instructor compensation workflows.

5.4 Virtual Laboratory and Infrastructure Domain

VMSession Class: The `VMSession` class represents an engineer's temporary remote laboratory instance. It belongs to a user, server, and compute node, and connects to attached devices and hardware queues. It is the main bridge between educational access and infrastructure provisioning.

ProxmoxServer and ProxmoxNode Classes: `ProxmoxServer` models the virtualization cluster endpoint, while `ProxmoxNode` models the compute node hosted under that server. These classes represent capacity, status, and allocation context required for provisioning VM sessions.

TrainingUnitVMAssignment Class: The `TrainingUnitVMAssignment` class links pedagogical content to technical infrastructure by allowing a training unit to request and receive an approved virtual machine environment. It connects teacher requests to administrator approval and provides a controlled way to embed hands-on lab work inside a training path.

GatewayNode, UsbDevice, Camera, and Reservation Classes: `GatewayNode` models the hardware gateway responsible for exposing physical resources. `UsbDevice` represents a manageable peripheral, while `Camera` represents a visual sensor that may also be associated with a robot or USB device. Reservation behavior is represented through `Reservation`, `UsbDeviceReservation`, and `CameraReservation`. These classes support availability planning, conflicts, approval, activation, and maintenance-aware control.

CameraSessionControl and Queueing Classes: `CameraSessionControl` represents temporary exclusive control of a camera inside a VM session. `UsbDeviceQueue` models waiting behavior when multiple users contend for limited hardware resources. These classes

express concurrency control within the remote laboratory domain. Figure 1.30 presents the complete class diagram of the IoT-REAP platform, organized into the domain packages described above: User and Access, Learning Content, Commerce and Revenue, and Virtual Laboratory and Infrastructure.

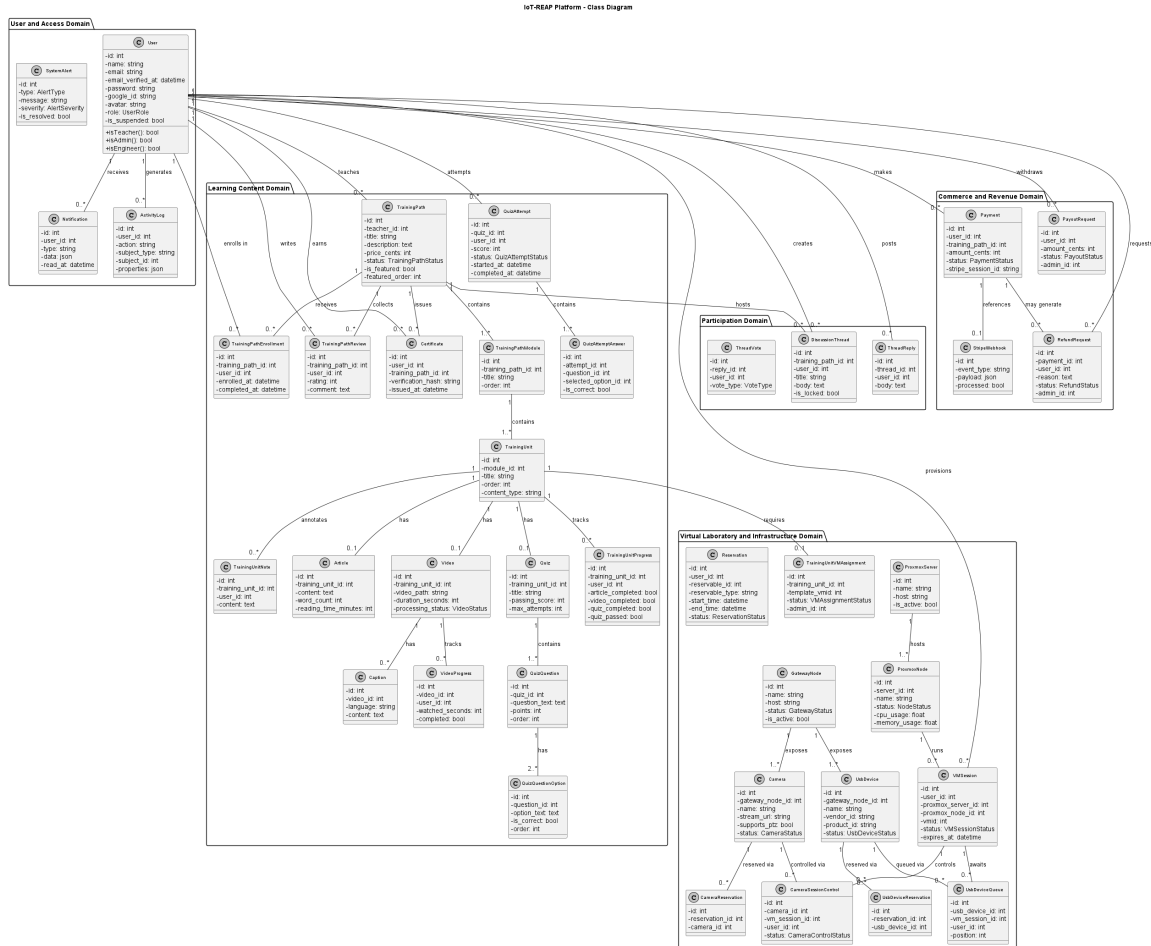


Figure 1.30: Class Diagram of the IoT-REAP Platform

6 Physical Data Model

The physical data model follows the domain structure exposed by the Eloquent models and route organization. At database level, the platform is centered on the `users` table, which is referenced by multiple transactional and educational tables. From this anchor, the model expands into five major relational clusters.

6.1 Learning and Authoring Tables

The training subsystem is organized around `training_paths`, `training_path_modules`, and `training_units`. The relationship is hierarchical:

one training path contains many modules, and one module contains many training units. Additional one-to-one or one-to-many tables extend training units with content and pedagogy, including articles, videos, captions, quizzes, quiz_questions, quiz_question_options, quiz_attempts, and quiz_attempt_answers. Progress-oriented tables such as training_unit_progress, video_progress, and training_unit_notes preserve learner-state evolution over time.

6.2 Participation and Certification Tables

Collaborative learning is represented by discussion_threads, thread_replies, and thread_votes. Satisfaction and reputation are represented by training_path_reviews. Enrollment membership is represented by training_path_enrollments. Completion evidence is represented by certificates, which reference both user and training path and expose verifiable hashes for public consultation.

6.3 Payment and Revenue Tables

Commercial activity is represented by payments, refund_requests, payout_requests, and stripe_webhooks. These tables preserve financial state transitions, external payment event traceability, and instructor revenue workflows. The data model ensures that one payment belongs to one user and one training path, while a payment may receive multiple related refund records depending on business logic.

6.4 Virtual Machine and Infrastructure Tables

Remote laboratory persistence is built around vm_sessions, proxmox_servers, proxmox_nodes, training_unit_vm_assignments, and user-specific connection preference tables such as guacamole_connection_preferences and user_vm_connection_default_profiles. This cluster stores compute allocation, session duration, infrastructure ownership, and preferred access settings.

6.5 Hardware and Reservation Tables

The IoT and device-control cluster includes gateway_nodes, usb_devices, cameras, camera_session_controls, reservations, usb_device_reservations, camera_reservations, and usb_device_queues. These tables model the

availability, attachment, queueing, maintenance, and scheduling logic required to safely expose real laboratory resources to remote users.

6.6 Governance and Monitoring Tables

Operational governance is supported by notifications, activity_logs, system_alerts, node_credentials_logs, and derived or statistical tables such as daily_training_path_stats. Together, these structures provide auditability, accountability, administrative observability, and monitoring support for a platform that mixes education with infrastructure control.

Overall, the physical data model reflects a normalized but highly interconnected design. Foreign keys enforce ownership and navigation between users, content, payments, reservations, and infrastructure entities. This organization makes the database suitable for both transactional consistency and operational extensibility. Figure 1.31 presents the entity-relationship diagram of the IoT-REAP physical data model, showing the major tables, their attributes, and the foreign key relationships across the six relational clusters.

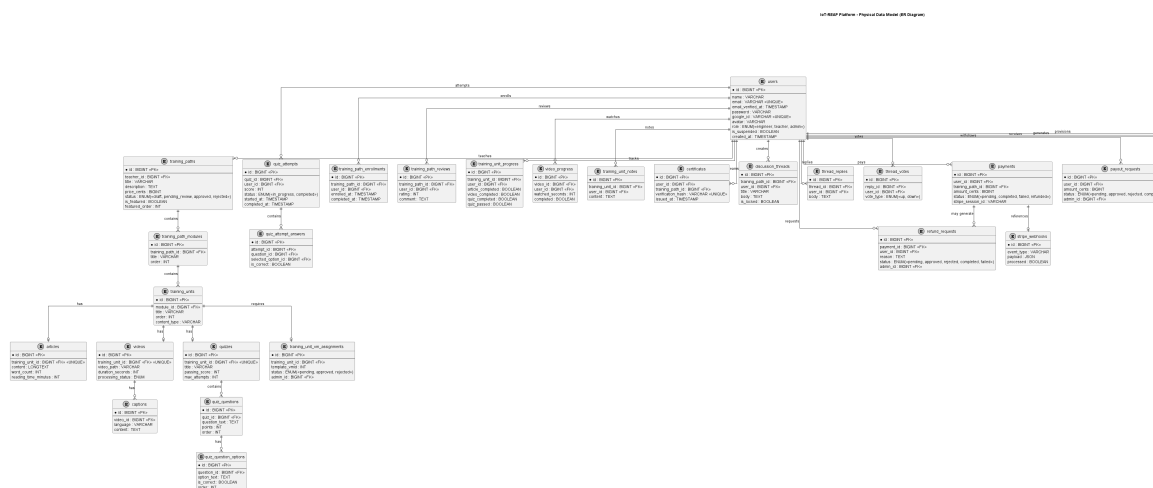


Figure 1.31: Entity-Relationship Diagram of the IoT-REAP Physical Data Model

7 Component Architecture Diagram

The component architecture of IoT-REAP reflects the separation of concerns between the frontend presentation layer, the backend application layer, and the external infrastructure services. The platform is built as a monolithic Laravel application with an Inertia.js-bridged React frontend, communicating with Proxmox VE for virtualization, Apache Guacamole for remote desktop proxying, an MQTT broker for robot telemetry and control, Stripe for payment processing, and USB/IP gateways for hardware device access.

The frontend layer consists of React components organized into pages, shared components, custom hooks, Zustand state stores, and typed API modules. The backend layer follows the request lifecycle pattern: FormRequest validation, thin Controllers, Service classes containing business logic, Repository classes for data access, Eloquent Models for domain representation, and API Resources for response shaping. Background processing is handled through Laravel Queues with jobs for VM provisioning, video processing, and cleanup operations.

The infrastructure layer includes Proxmox VE clusters for compute virtualization, Apache Guacamole for protocol-adaptive remote access (RDP, VNC, SSH), Mosquitto MQTT broker for real-time robot communication, Stripe for payment and refund processing, and USB/IP gateway nodes for physical device exposure. Laravel Echo and Pusher provide real-time event broadcasting for notifications and live updates.

Figure 1.32 presents the component architecture diagram, showing the major subsystems and their communication pathways.

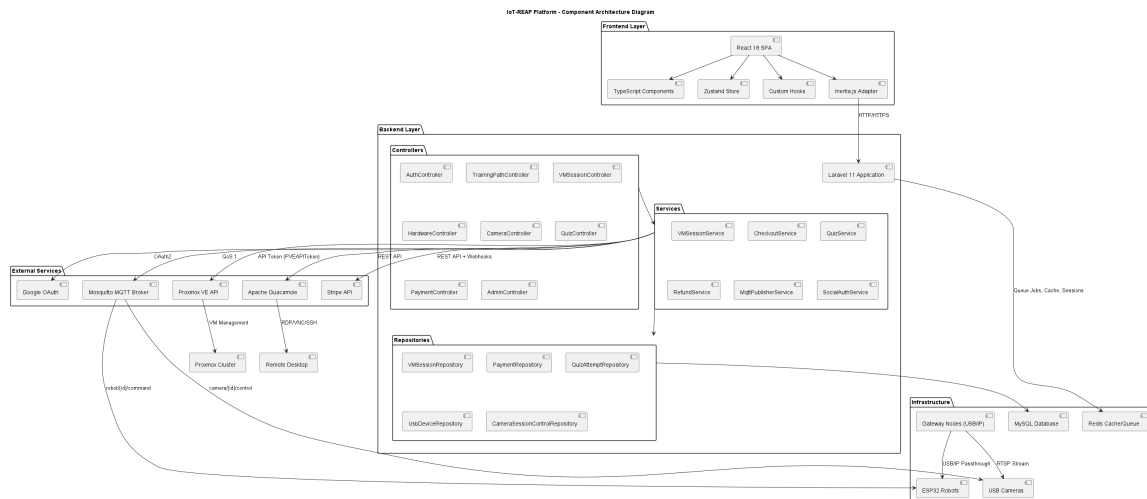


Figure 1.32: Component Architecture Diagram of the IoT-REAP Platform

8 Deployment Diagram

The deployment of IoT-REAP spans multiple physical and virtual nodes, reflecting the hybrid nature of an educational platform that integrates cloud-hosted application services with on-premises laboratory infrastructure. The deployment architecture consists of the following principal nodes.

The **Web Application Server** hosts the Laravel application, the React frontend served through Inertia.js, the Nginx reverse proxy, and the PHP-FPM process manager. This node also runs the MySQL database, the Redis cache and session store, and the Laravel queue worker processes. The Mosquitto MQTT broker may be co-located on this server or deployed on a separate node depending on traffic requirements.

The **Proxmox VE Cluster** comprises one or more physical servers running the Proxmox hypervisor. Each cluster node hosts virtual machines that are provisioned on demand for engineer lab sessions. The cluster is accessed through the Proxmox API using token-based authentication.

The **Apache Guacamole Gateway** runs as a separate service that proxies remote desktop protocols (RDP, VNC, SSH) to the engineer's browser. It connects to the provisioned VMs through the internal network and exposes a WebSocket-based client interface.

The **USB/IP Gateway Nodes** are physical machines running the USB/IP kernel module and custom gateway management software. Each gateway node exposes attached USB devices and cameras to the virtual machines through IP-based USB forwarding. The gateways register with the main application and report device availability and health status.

The **Stripe Payment Service** operates as an external SaaS endpoint, receiving checkout sessions and webhook callbacks over HTTPS. It is not deployed within the platform infrastructure but is integrated through the Stripe SDK and webhook signature verification.

Figure 1.33 presents the deployment diagram, showing the physical nodes, the software artifacts deployed on each node, and the communication protocols between them.

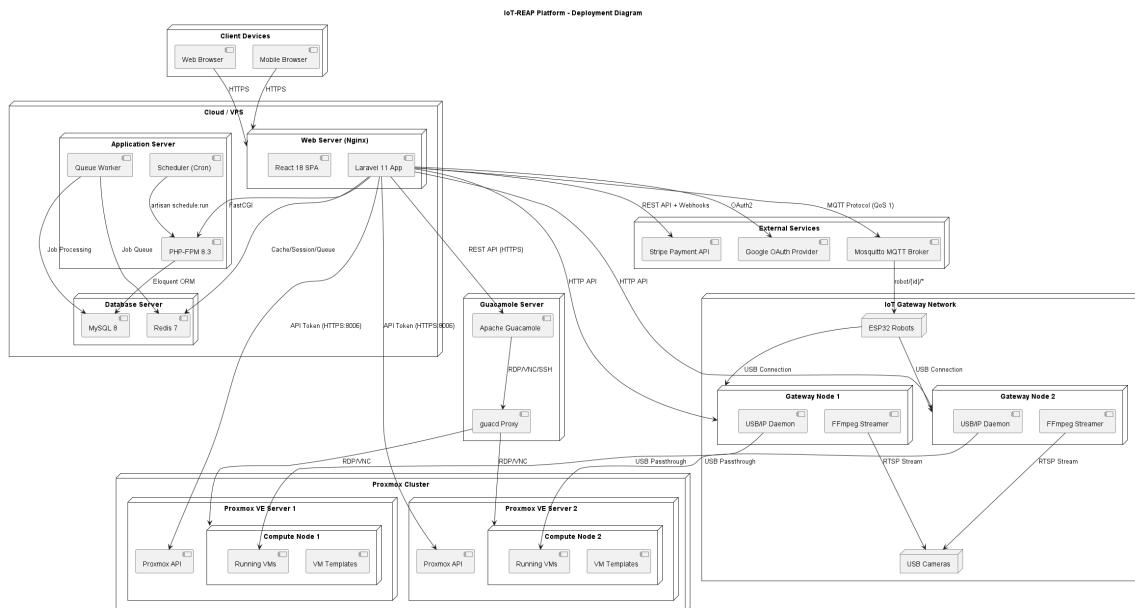


Figure 1.33: Deployment Diagram of the IoT-REAP Platform

Conclusion

This chapter presented the conceptual analysis of the current IoT REAP platform. The rewrite replaced the outdated assessment-oriented description with a model that corresponds to the implemented system: a role-based learning and remote laboratory platform integrating

public discovery, course authoring, learner progression, payment workflows, virtual machine provisioning, device and camera reservations, and administrative governance. The actor analysis, requirements specification, use case refinements, sequence and activity descriptions, class model, physical data model, component architecture, and deployment diagram together provide a coherent conceptual foundation for the technical realization discussed in the following chapters.